



Bharati Vidyapeeth's
College of Engineering, New Delhi

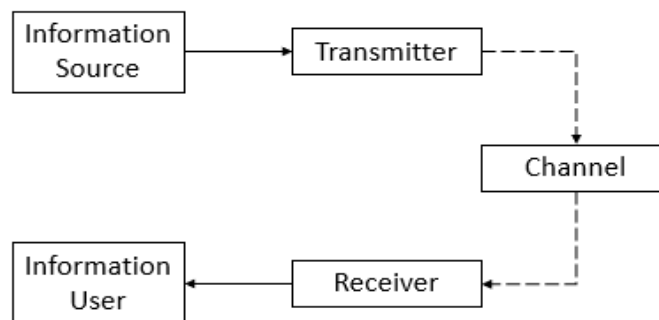
ECC-209

Analog Communication

Unit – 1

Subject Introduction

A typical communication system starts with a source of information of interest that we want to send somewhere. Normally, it is required to be converted to electrical form using a transducer. After its electrical conversion, we use an equipment called transmitter, the work of which is to prepare the signal for the physical medium which it is to be sent over. In essence, the job of the transmitter is to match the properties of the information signal to the properties of the physical medium over which it is to be sent so that efficient communication is possible. An abstraction of all kinds of physical mediums through which transmission can take place is termed as a channel. At the other end or destination, there is a receiver the job of which is to interpret the received information in a way that is usable to the user. Hence, in a communication system, there are five major blocks connected as shown.





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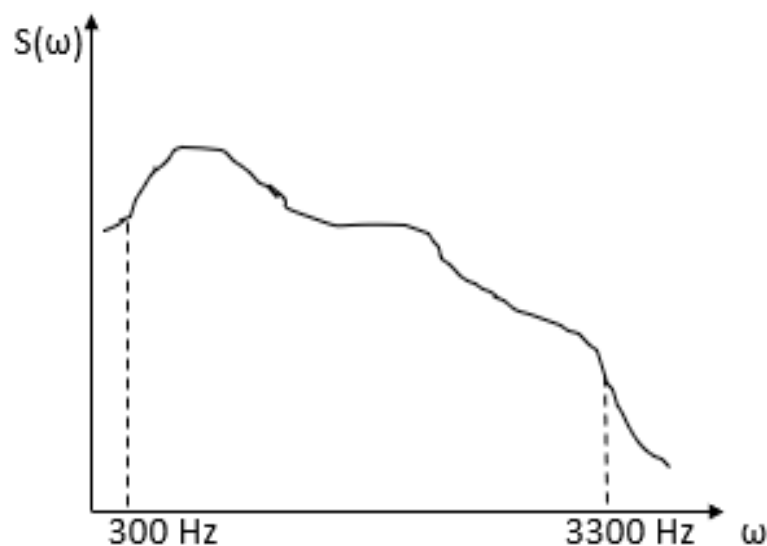
Information Sources

Communication can use a wide range of sources of information based upon the type of data being transmitted. Voice / Speech signals make up the bulk of communication and is easily the most common form of information. TV and Fax involve communication using moving and still pictures respectively. Data signals are commonly used in the communication of computers. There might be many other signals or information sources, but these three broad categories roughly cover all useful class of sources.

One of the most basic frameworks needed in the discussion of deterministic signals used as information sources is the spectrum of the signal. In its basic sense, the spectrum can be understood as the Fourier Transform of the signal. As is known, the FT of a real signal is (normally) a complex function, meaning that it has both magnitude as well as phase components varying with respect to frequency, called Magnitude and Phase spectrums respectively. Following is a brief discussion on the afore mentioned types of signals in context of their spectrums.

Information Sources

A typical spectrum of human speech signals can be drawn as shown here. As can be seen in the graph, there might be some peaks, but the maximum amount of the intelligent data used in the speech/ voice communication can be seen to lie in the frequency range of 300Hz to 3300Hz. Including some guard band, we may infer that the bandwidth (maximum possible relevant frequency in the spectrum) of a speech signal is 4KHz. It is noteworthy though, that the actual spectrum of the voice signal may go up to 7 – 8 KHZ. Thus, for studio quality voice processing, the bandwidth must be kept of the same order. In telephonic communication, however, the above mentioned 4KHz bandwidth is considered adequate.



Frequency Spectrum of typical Speech signal



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Picture signals

As discussed earlier, the picture signal can be of two forms.

Moving Pictures/ Television/ Video transmission

While transmitting picture signals with fast changing characteristics, interpreted as moving pictures, care must be taken that we don't lose the essence of motion itself. We first need to convert the special picture signal (2D in nature) into an electrical form (1D time-varying signal). This is done using a picture tube and the output is scanned (horizontally) line wise. This scan must be fast enough to retain the essence of motion of the pictures. It can hence be interpreted, that this fast variation required would mean larger bandwidth requirement. Typically, the TV signals are seen to have a bandwidth of 4.5MHz to 5MHz.

Still Pictures/Fax/Photo Transmission

In a fax signal or a signal transmitting a single image, speed of transmission is not a concern. The image in the form of electrical signals can be scanned at any speed convenient to us based upon the available hardware. Thus, Fax signals are possible to transit over general telephone lines which are designed to operate on a bandwidth much smaller than that of video signals.



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Data signals

This is a very different type of information in the sense that it is digital in nature in comparison to the other sources discussed so far, which were analog. It must be noted though that the channel or the physical medium through which transmission takes place is still analog in nature, hence in the context of effective transmission, it doesn't make much difference. What is interesting to note though is that the bandwidth of transmission of data signals will depend upon the rate at which data transmission is needed (data rate in bits/s). Thus, a fast data rate system requires larger bandwidth in comparison to a slow data rate system that requires low bandwidth. Consider the example of mobile networks evolving from 2G to 3G to 4G and so on. The idea is to increase the bandwidth allocation per user to increase the rate at which transmission can take place.



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Transmitter

Basic function of the transmitter is to match the signal characteristics of the output of the information source with the characteristics of the channel. Let us try to understand this using an example. Say the transmitter needs to transmit data on to free space. For such transmission, one requires an appropriate antenna, the design of which depends upon the frequency of operation. It is a known fact in electromagnetics, that the height of an antenna transmitting EM waves must be of the order of the wavelength of the wave. Considering a voice signal, the maximum frequency of transmission is kept at 4KHz. Thus, the wavelength of such a signal is nearly 75km! Obviously, an antenna of comparable height will be impossible to design efficiently. Hence, the information source output is not matched to the channel of transmission. In this case, the job of the transmitter would be to embed the information signal on to another signal for which the antenna height requirement is much smaller. It can easily be deduced that such signals will be of very high frequency. As these signals carry the information signal in some form, these are known as carrier signals. Typically, the carrier wave is a pure sinusoid of high frequency. The process of embedding the information on to a carrier is called modulation. Thus, ease of radiation is one of the reasons why modulation is needed. There are some other needs which may also be studied as advantages and will be discussed a little later.



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Channel

A channel is basically a physical medium. Some examples are cables, optical fibre, free space, etc. Another way of looking at a channel is as an abstraction. This means that channel is to be modelled in an abstract fashion. Thus, all the effects of the transmitter, receiver and physical medium that are undesirable or unexpected are modelled as a block called channel. These undesired effects can be simply called noise. Hence, to say that noise exists in a communication system simply means that noise exists in the channel.

Noise in Communication Systems

Broadly, noise sources can be classified into two categories: Internal noise and External Noise. External noises are attributed to sources outside the circuit. Natural sources may include lightening, atmospheric noises, cosmic EM waves, etc. Man-made sources may include EM distortions generated by electrical lines, commutator switches in motors, ignition noises in automobiles, etc. Along with this, other Radio frequency interferences have become more prominent with the advent of multiuser communication over similar frequency spectrums. Multipath fading is also a common issue in wireless communication. An in-depth discussion of different forms of noises will be done later in this course.

Propagation in a Communication Channel

The various methods of propagation of signals in a communication channel depends upon the type of channel itself. Based on this criterion, the classification can be done as follows:

EM wave propagation channel – Free space channel

Guided EM wave propagation channel

Optical Channels

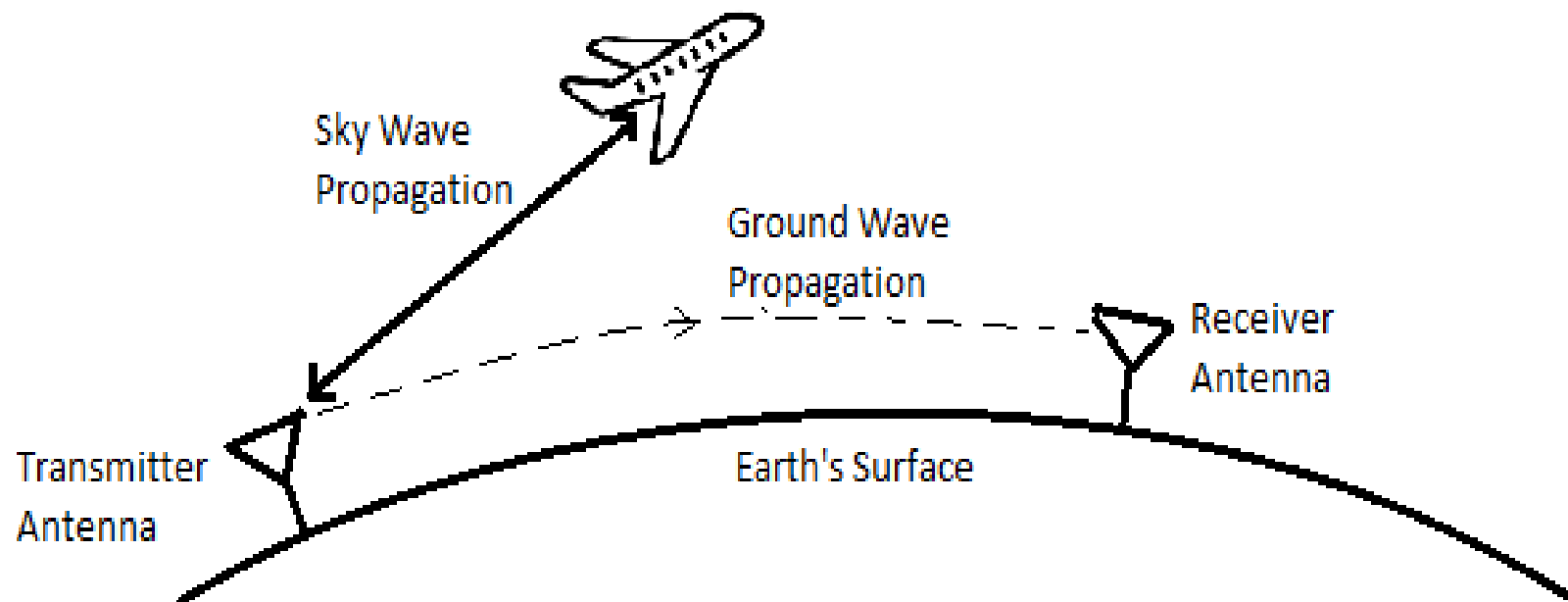
As such, pure free space communication cannot take place in the earth's atmosphere. But still wireless communication can be considered free space communication under specific assumptions.

Propagation in a Communication Channel

As is known, a medium bounded on one side by a conductor gives rise to guided waves. In the case shown here, transmitter antenna is sending an EM wave to a nearby receiver. In such a case, the Earth's surface would act as a conductor and the curvature of Earth guides the wave. This is called the ground wave propagation and classified as a guided transmission (type b). The limitation of such a system is that it can only be used for low frequency transmission, since for higher frequencies, the attenuation due to Earth's surface increases significantly. Therefore, ground wave propagation is well suited for lower frequencies.

Consider now, that the communication needs to take place between a transmitter at the Earth's surface and an Aeroplane. Such communication can only take place in Line of Sight (LoS). Since this involved transmission outwards from the Earth's surface, for all practical purposes it can be termed as free space communication, although a more suited name is Sky Wave Communication. This transmission can work for higher frequencies as well.

Propagation in a Communication Channel



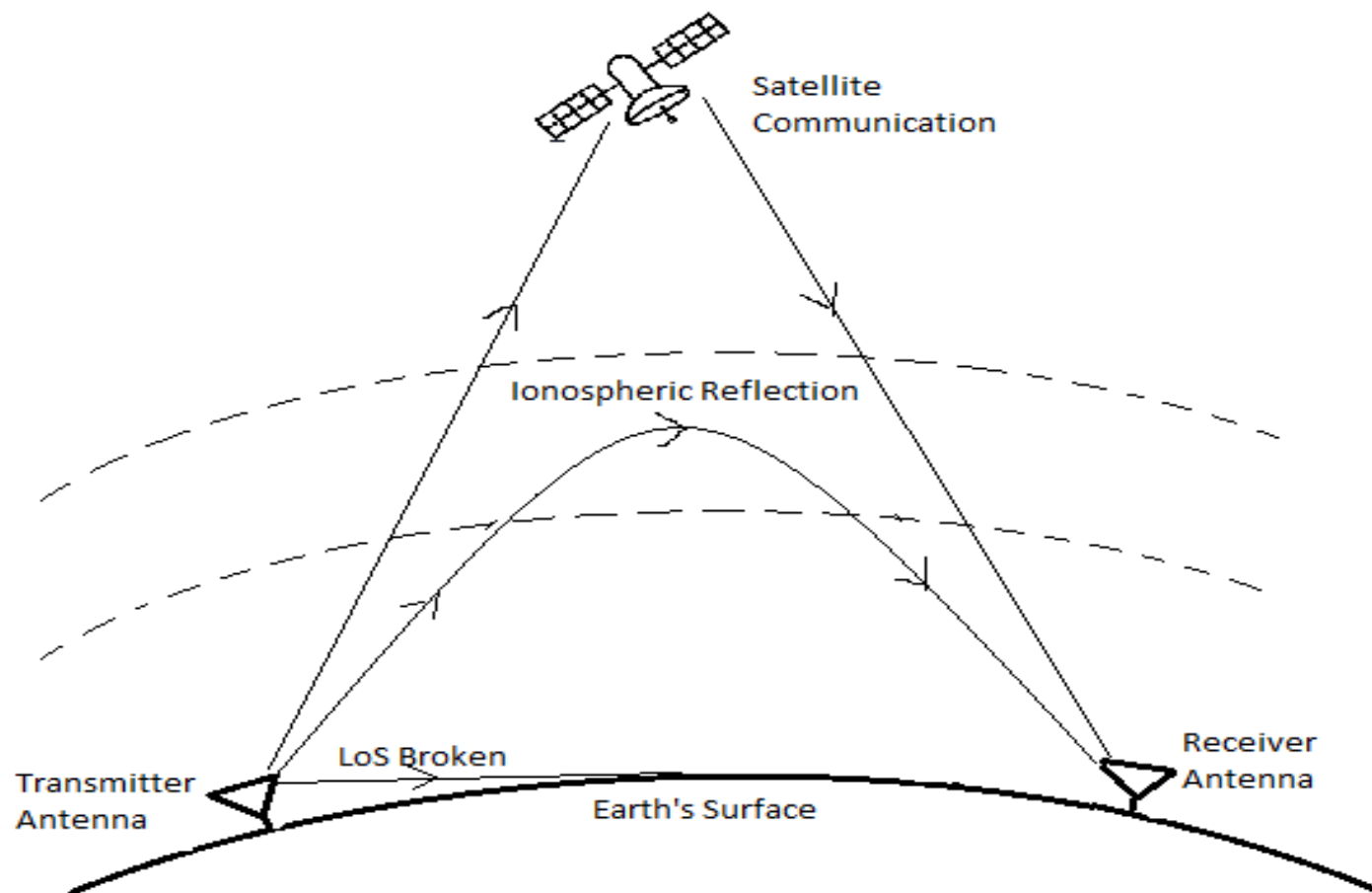
Propagation in a Communication Channel

LoS communication is also possible between stations on the Earth's surface. This is, though, limited by the distance between the stations. Due to the curvature of the Earth, there will be a point beyond which the signal will be obstructed by the surface itself.

The way out of this problem is to transmit the wave skywards and somehow have it reflected to the destination. One way of doing so is making use of a natural phenomenon. In the Earth's atmosphere, there is a layer, called Ionosphere, which is made up of charged ions and acts as a passive reflector. The limitation of this is that the ionosphere is capable of reflecting frequencies up to 30MHz. Higher frequencies will escape the ionosphere and move into free space.

For higher frequencies ($>30\text{MHz}$), we need to have a reflecting satellite in an orbit around the planet. If that satellite only reflects the wave, it is called a passive satellite. In case it receives and retransmits the wave, it is called an active satellite.

Propagation in a Communication Channel



Propagation in a Communication Channel

Among the guided wave channels, some more examples that can be quoted are:

- a) Cables and Wire pairs: Depending upon the lower frequencies or higher frequencies, a pair of wires may be modelled as a lumped circuit or transmission lines respectively. In each case, they are used as guided media to transmit EM waves.
- b) Waveguides: They are essentially pipes (cylindrical or rectangular) used to transfer EM waves from transmitter circuit to the antenna.

Optical Channels

Optical communication is possible using Fibre Optics as well as Free Space Optics. Fibre optics is a cylindrical waveguide used for transmission of optical signals, which in turn are essentially EM waves. Free Space Optics is typically like a Free Space EM communication. The main difference is that the information is transmitted in the form of highly focussed beam of light to enable transmission from point A to B. Mainly, Free Space Optics is used in sky wave communication, but it is also finding many terrestrial applications as well.

What Signals have Fourier Transforms?

(1) $x(t)$ can be of infinite duration, but must satisfy:

a) Finite energy $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$

In this case, there is *zero* energy in the error

$$e(t) = x(t) - \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{Then} \quad \int_{-\infty}^{\infty} |e(t)|^2 dt = 0$$

b) Dirichlet conditions (including $\int_{-\infty}^{\infty} |x(t)| dt < \infty$)

(i) $\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = x(t)$ at points of continuity

(ii) $\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = \text{midpoint at discontinuity}$

(iii) Gibb's phenomenon

c) By allowing impulses in $x(t)$ or in $X(j\omega)$, we can represent even *more* signals

Fourier Transform Examples

IMPULSES

(a) $x(t) = \delta(t)$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

$\Downarrow$

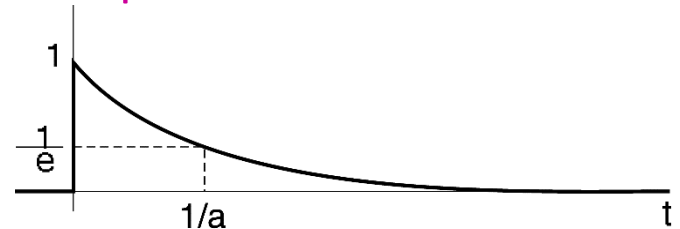
$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega$$

(b) $x(t) = \delta(t - t_0)$

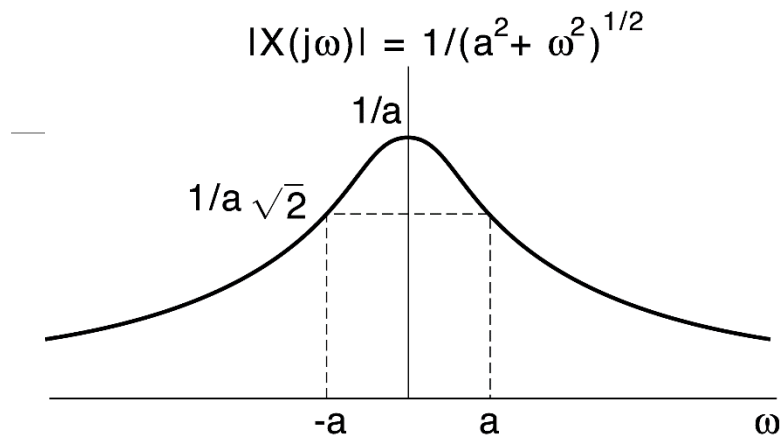
$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt \\ &= e^{-j\omega t_0} \end{aligned}$$

Fourier Transform of Right-Sided Exponential

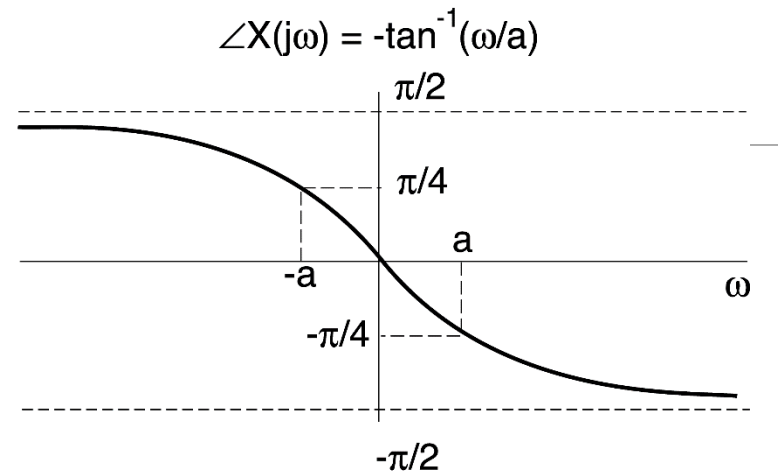
$$x(t) = e^{-at}u(t), a > 0$$



$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = \int_0^{\infty} \underbrace{e^{-at}e^{-j\omega t}}_{e^{-(a+j\omega)t}} dt \\ &= -\left(\frac{1}{a+j\omega}\right) e^{-(a+j\omega)t} \bigg|_0^{\infty} = \frac{1}{a+j\omega} \end{aligned}$$

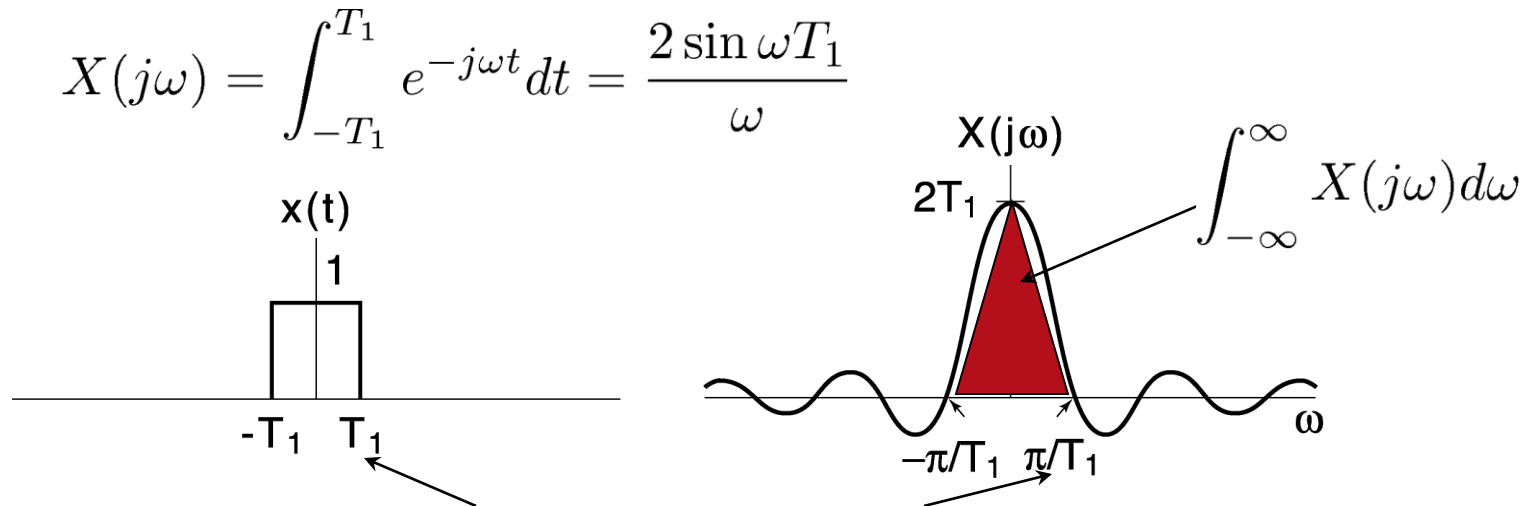


Even symmetry



Odd symmetry

Fourier Transform of square pulse



Note the inverse relation between the two widths $\Rightarrow$ Uncertainty principle

Useful facts about CTFT's

$$X(0) = \int_{-\infty}^{\infty} x(t) dt \quad \text{Example above: } \int_{-\infty}^{\infty} x(t) dt = 2T_1 = X(0)$$

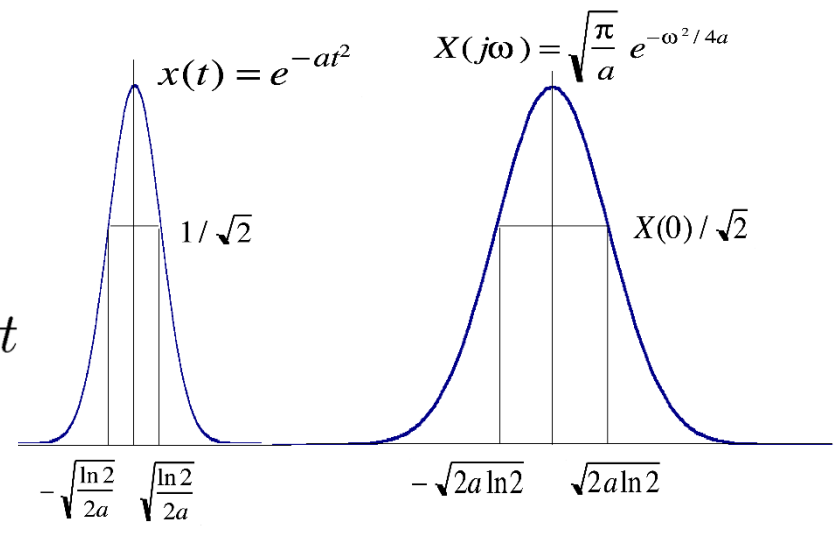
$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) d\omega$$

$$\begin{aligned} \text{Ex. above: } x(0) &= 1 = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) d\omega = \frac{1}{2\pi} \cdot \frac{1}{2} \left(2T_1 \times \frac{2\pi}{T_1} \right) \\ &= \frac{1}{2\pi} \times (\text{Area of the triangle}) \end{aligned}$$

Fourier Transform of a Gaussian

$x(t) = e^{-at^2}$ — A Gaussian, important in probability, optics, etc.

$$\begin{aligned}
 & X(j\omega) \\
 = & \int_{-\infty}^{\infty} e^{-at^2} e^{-j\omega t} dt \\
 = & \int_{-\infty}^{\infty} e^{-a \left[t^2 + j \frac{\omega}{a} t + \left(\frac{j\omega}{2a} \right)^2 \right] + a \left(\frac{j\omega}{2a} \right)^2} dt \\
 = & \underbrace{\left[\int_{-\infty}^{\infty} e^{-a \left(t + \frac{j\omega}{2a} \right)^2} dt \right]}_{\sqrt{\pi}/a} \cdot e^{-\frac{\omega^2}{4a}} \\
 = & \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}
 \end{aligned}$$



(Pulse width in t) • (Pulse width in ω)
 $\Rightarrow \Delta t \cdot \Delta \omega \sim (1/a^{1/2}) \cdot (a^{1/2}) = 1$

CT Fourier Transforms of Periodic Signals

Suppose

$$X(j\omega) = \delta(\omega - \omega_0)$$

$\Downarrow$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t}$$

That is

$$\boxed{e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)}$$

More generally

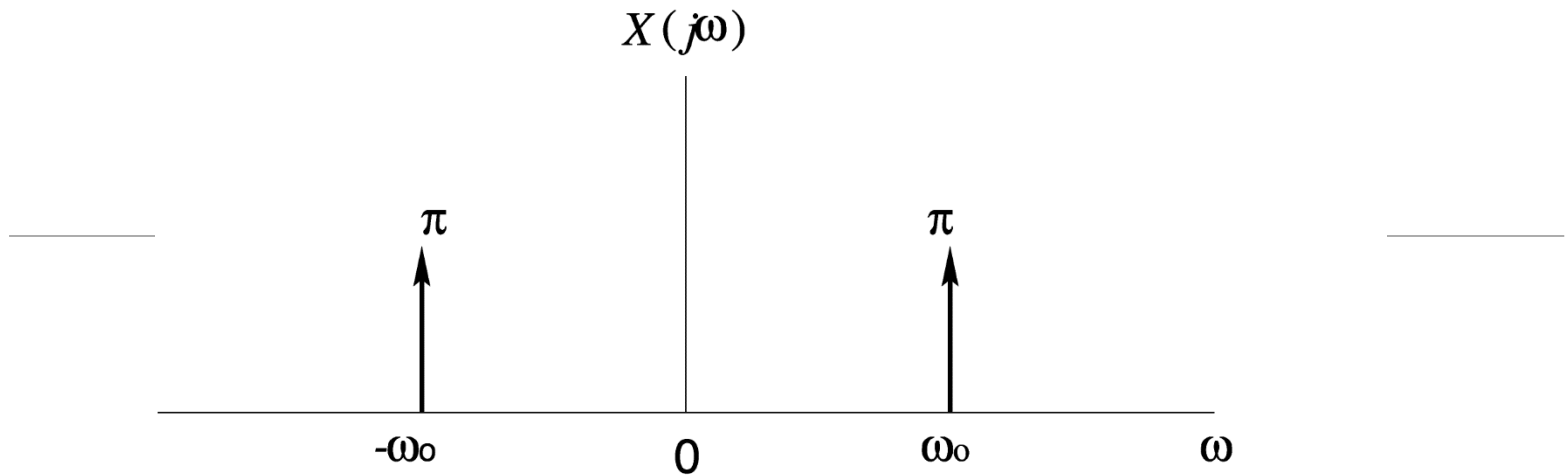
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \leftrightarrow X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

Fourier Transform of Cosine

$$x(t) = \cos \omega_0 t = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$

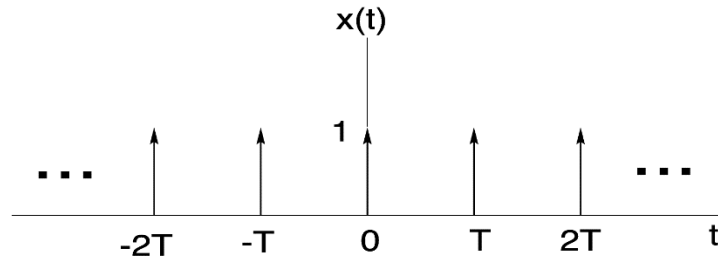
$\Updownarrow$

$$X(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$



Impulse Train (Sampling Function)

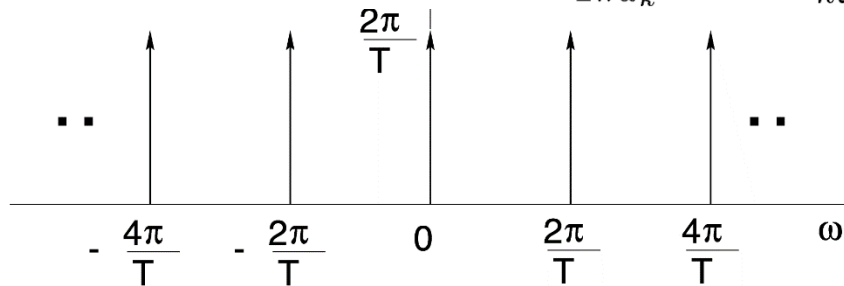
$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$



$$x(t) \leftrightarrow a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T}$$

$\Downarrow$

$$X(j\omega) = \sum_{n=-\infty}^{\infty} \underbrace{\frac{2\pi}{T}}_{2\pi a_k} \delta\left(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_0}\right)$$



Note: (period in t) T
 $\Leftrightarrow$ (period in ω) $2\pi/T$

Properties of the CT Fourier Transform

1) **Linearity** $ax(t) + by(t) \leftrightarrow aX(j\omega) + bY(j\omega)$

2) **Time Shifting** $x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$

Proof:
$$\int_{-\infty}^{\infty} \underbrace{x(t - t_0)}_{t'} e^{-j\omega t} dt = e^{-j\omega t_0} \underbrace{\int_{-\infty}^{\infty} x(t') e^{-j\omega t'} dt'}_{X(j\omega)}$$

FT magnitude unchanged

$$|e^{-j\omega t_0} X(j\omega)| = |X(j\omega)|$$

Linear change in FT phase

$$\angle(e^{-j\omega t_0} X(j\omega)) = \angle X(j\omega) - \omega t_0$$

Properties (continued)

3) Conjugate Symmetry

$$x(t) \text{ real} \leftrightarrow X(-j\omega) = X^*(j\omega)$$

$\Downarrow$

$$|X(-j\omega)| = |X(j\omega)| \quad \text{Even}$$

$$\angle X(-j\omega) = -\angle X(j\omega) \quad \text{Odd}$$

$$\text{Or} \quad \text{Re}\{X(-j\omega)\} = \text{Re}\{X(j\omega)\} \quad \text{Even}$$

$$\text{Im}\{X(-j\omega)\} = -\text{Im}\{X(j\omega)\} \quad \text{Odd}$$

When $x(t)$ is real (all the physically measurable signals are *real*), the negative frequency components do *not* carry any additional information beyond the positive frequency components: $\omega \geq 0$ will be sufficient.

More Properties

4) **Time-Scaling** $x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$

$$\Downarrow a = -1$$

$$x(-t) \longleftrightarrow X(-j\omega)$$

E.g. $a > 1 \rightarrow at > t$
compressed in time $\Leftrightarrow$
stretched in frequency

$$\Downarrow$$

a) $x(t)$ real and even $x(t) = x(-t) = x^*(t)$

$$\Rightarrow X(j\omega) = X(-j\omega) = X^*(j\omega) - \text{Real \& even}$$

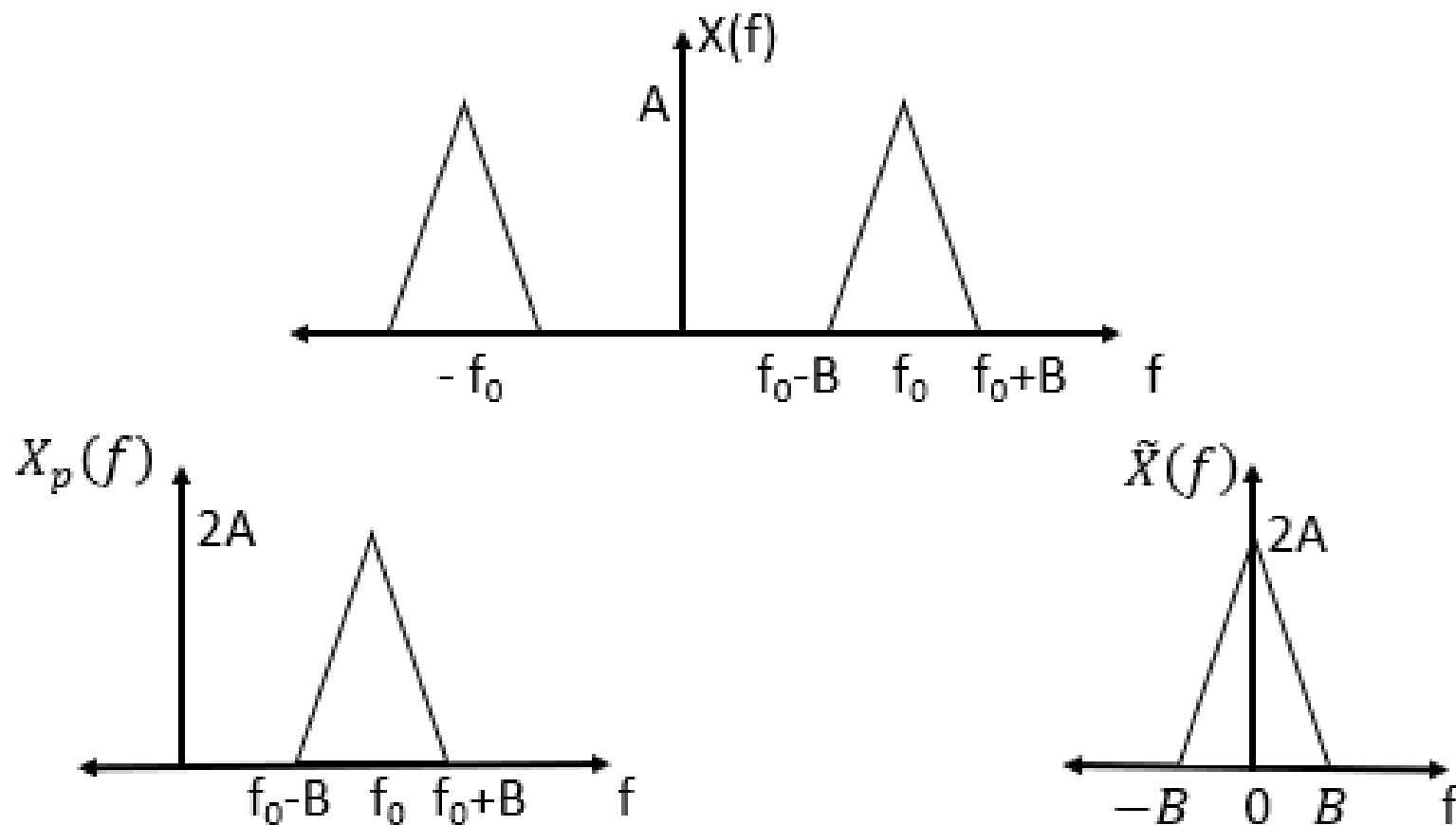
b) $x(t)$ real and odd $x(t) = -x(-t) = x^*(t)$

$$\Rightarrow X(j\omega) = -X(j\omega) = -X^*(j\omega) - \text{Purely imaginary}$$

c) $X(j\omega) = \text{Re}\{X(j\omega)\} + j\text{Im}\{X(j\omega)\}$

$$\text{For real } x(t) = \underset{\uparrow}{\text{Ev}\{x(t)\}} + \underset{\uparrow}{j\text{Od}\{x(t)\}}$$

Analytic Representation of Band Pass Signals



Analytic Representation of Band Pass Signals

What we may deduce from this is that $|x_p(t)| = |\tilde{x}(t)|$.

Further, $x(t) = \text{Re}\{x_p(t)\}$. We can imagine $\tilde{x}(t)$ to be a complex valued signal such that

$$\tilde{x}(t) = x_{Re}(t) + jx_{Im}(t)$$

Hence,

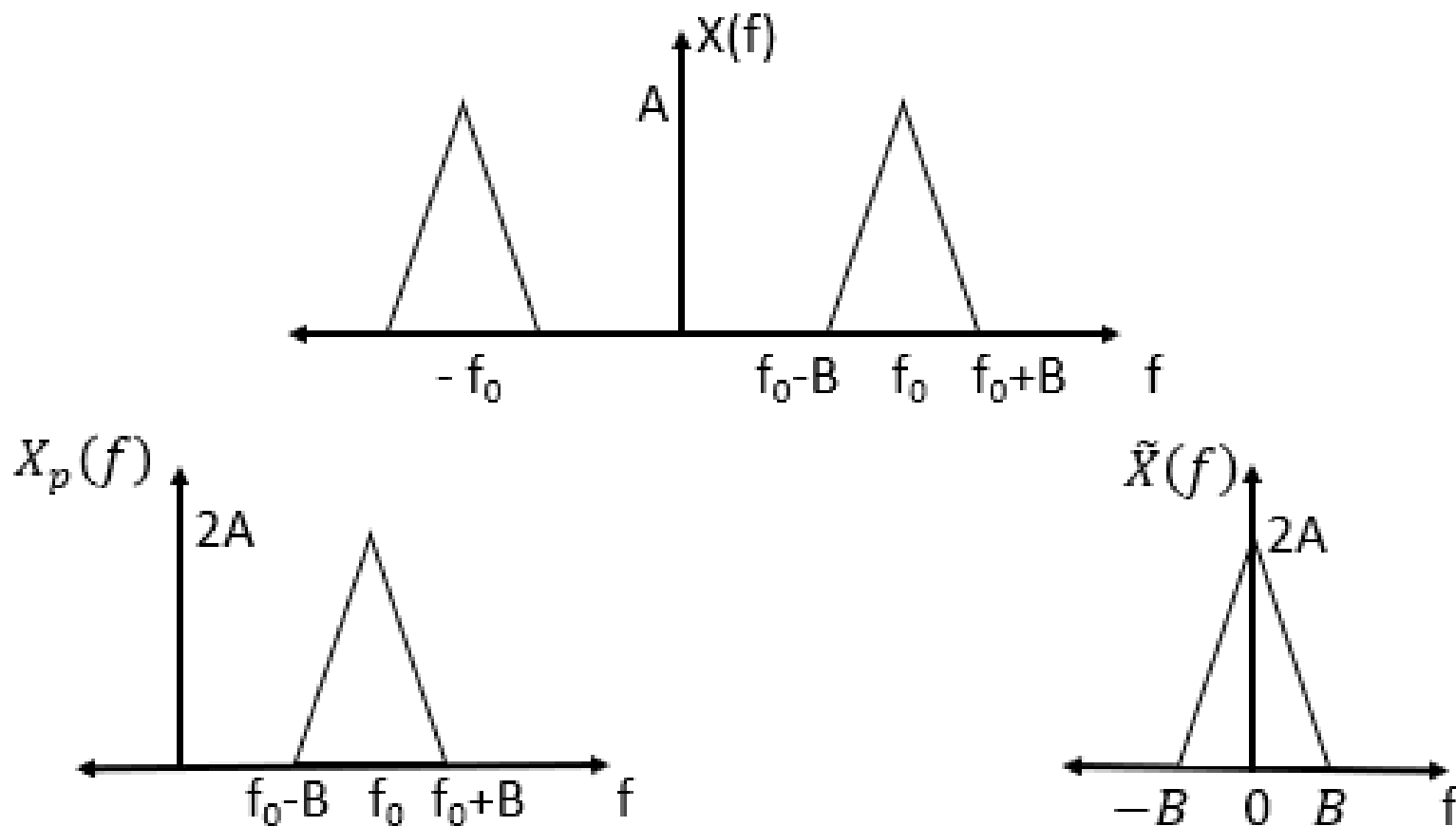
$$x(t) = x_{Re}(t) \cos 2\pi f_0 t - x_{Im}(t) \sin 2\pi f_0 t$$

which again re-iterates the fact that any bandpass signal can be represented as a product of a low pass (representative) term with a high frequency carrier. The value of $x(t)$ has two components, one associated with $\cos 2\pi f_0 t$ and another with $\sin 2\pi f_0 t$, thus one of them is in phase with the carrier and the other is at a quadrature phase difference with the carrier. For this reason, the component $x_{Re}(t)$ is also called the in-phase component $x_I(t)$ and the component $x_{Im}(t)$ is also called the quadrature component $x_Q(t)$. This representation, called the quadrature representation, makes the analysis of band pass signals very much convenient. Even the other representation

$$x(t) = \text{Re}\{\tilde{x}(t)e^{j2\pi f_0 t}\}$$

is equally graceful. This representation is called the complex envelop representation.

Analytic Representation of Band Pass Signals



Amplitude Modulation

A large number of information sources produce analog signals

- Analog signals can be modulated and transmitted directly, or
- They can be converted into digital data and transmitted using digital-modulation techniques

The notion of analog-to-digital conversion : Examined in detail in Chapter 7

Speech, music, images, and video are examples of analog signals

- Each of these signals is characterized by its bandwidth, dynamic range, and the nature of the signal
- Speech signals : Bandwidth of up to 4 kHz
- Audio and black-and-white video
 - The signal has just one component, which measures air pressure or light intensity
- Music signal : Bandwidth of 20 kHz
- Color video
 - The signal has four components, namely, the red, green, and blue color components, plus a fourth component for the intensity
 - In addition to the four video signals, an audio signal carries the audio information in Color-TV broadcasting
 - Video signals have a much higher bandwidth, about 6 MHz

INTRODUCTION TO MODULATION

The analog signal to be transmitted is denoted by $m(t)$

- Assumed to be a lowpass signal of bandwidth W
- $M(f) = 0$, for $|f| > W$

- The power content of this signal is denoted by $P_m = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |m(t)|^2 dt$

- The message signal $m(t)$ is transmitted through the communication channel by impressing it on a carrier signal of the form

$$c(t) = A_c \cos(2\pi f_c t + \phi_c)$$

- A_c : Carrier amplitude
- f_c : Carrier frequency
- ϕ_c : Carrier phase - The value of ϕ_c depends on the choice of the time origin
 - we assume that the time origin is chosen such that $\phi_c = 0$
- We say that the message signal $m(t)$ modulates the carrier signal $c(t)$ in either amplitude, frequency, or phase if after modulation, the amplitude, frequency, or phase of the signal become functions of the message signal
- Modulation converts the message signal $m(t)$ from lowpass to bandpass, in the neighborhood of the carrier frequency f_c .

AMPLITUDE MODULATION (AM)

In amplitude modulation, the message signal $m(t)$ is impressed on the amplitude of the carrier signal $c(t) = A_c \cos(2\pi f_c t)$

- This results in a sinusoidal signal whose amplitude is a function of the message signal $m(t)$
- There are several different ways of amplitude modulating the carrier signal by $m(t)$
- Each results in different spectral characteristics for the transmitted signal
- We will describe these methods, which are called
 - (a) Double sideband, suppressed-carrier AM (DSB-SC AM)
 - (b) Conventional double-sideband AM
 - (c) Single-sideband AM (SSB AM)
 - (d) Vestigial-sideband AM (VSB AM)

Double-Sideband Suppressed-Carrier AM

A double-sideband, suppressed-carrier (DSB-SC) AM signal is obtained by multiplying the message signal $m(t)$ with the carrier signal $c(t) = A_c \cos(2\pi f_c t)$

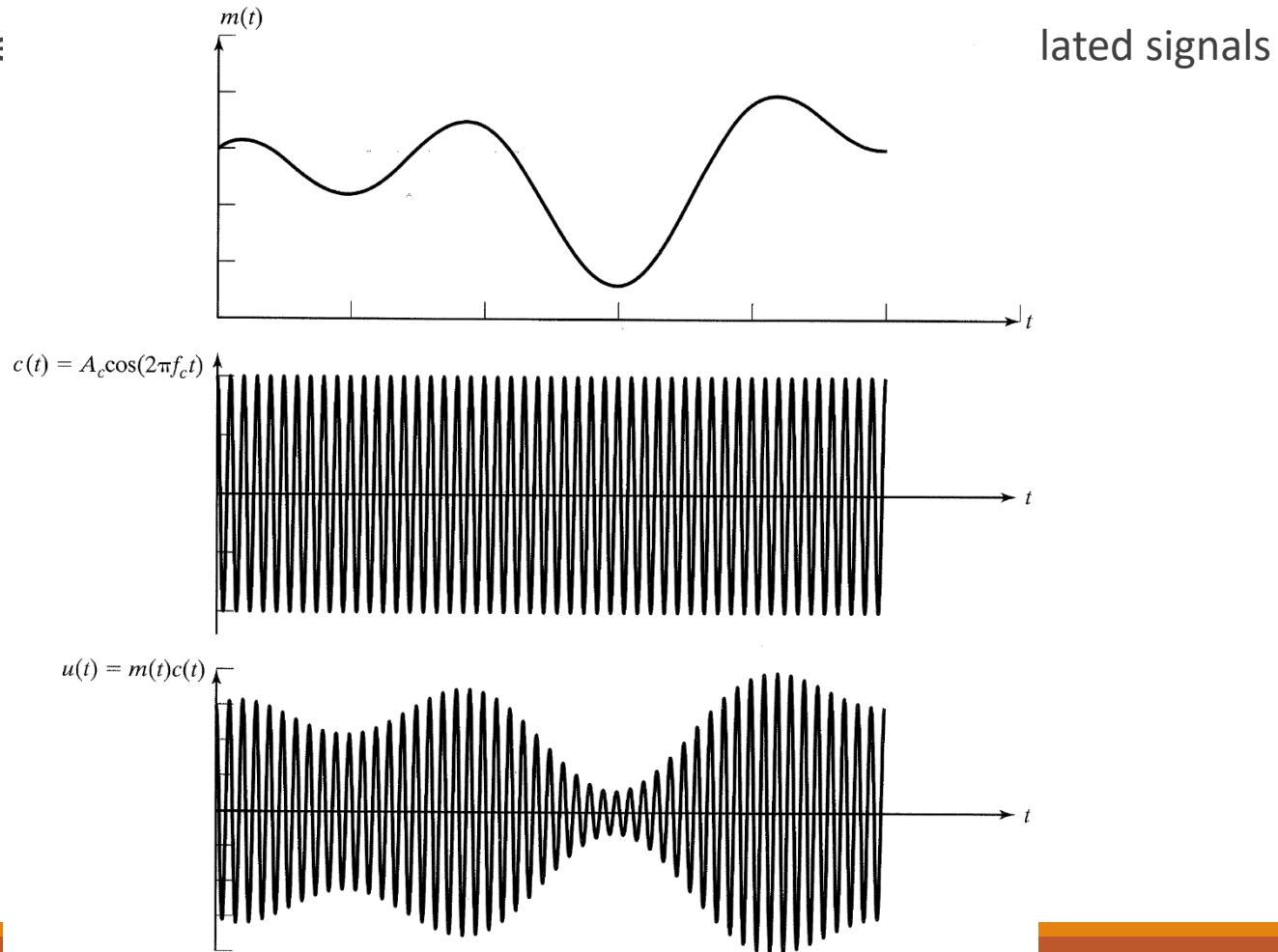
Amplitude-modulated signal

$$u(t) = m(t)c(t) = A_c m(t) \cos(2\pi f_c t)$$

- An example of the message signal $m(t)$, the carrier $c(t)$, and the modulated signal $u(t)$ are shown in Figure 3.1
- This figure shows that a relatively slowly varying message signal $m(t)$ is changed into a rapidly varying modulated signal $u(t)$, and due to its rapid changes with time, it contains higher frequency components
- At the same time, the modulated signal retains the main characteristics of the message signal; therefore, it can be used to retrieve the message signal at the receiver

Double-Sideband Suppressed-Carrier AM

Figure



Spectrum of the DSB-SC AM Signal

Spectrum of the modulated signal can be obtained by taking the FT of $u(t)$

$$U(f) = \frac{A_c}{2} [M(f - f_c) + M(f + f_c)]$$

Figure 3.2 illustrates the magnitude and phase spectra for $M(f)$ and $U(f)$

- The magnitude of the spectrum of the message signal $m(t)$ has been translated or shifted in frequency by an amount f_c
- The bandwidth occupancy, of the amplitude-modulated signal is $2W$, whereas the bandwidth of the message signal $m(t)$ is W
- The channel bandwidth required to transmit the modulated signal is $B_c = 2W$

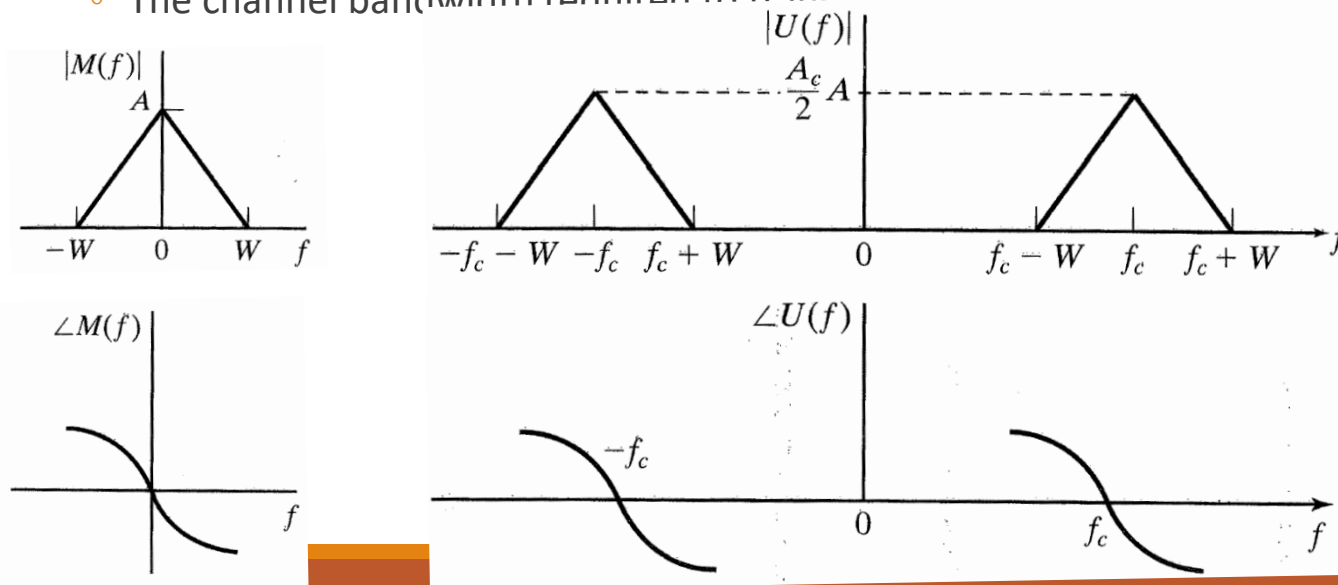


Figure 3.2
Magnitude and phase spectra of the message signal $m(t)$ and the DSB-AM modulated signal $u(t)$

Spectrum of the DSB-SC AM Signal

The frequency content of the modulated signal $u(t)$ in the frequency band

$|f| > f_c$ is called the *upper sideband* of $U(f)$

The frequency content in the frequency band $|f| < f_c$ is called the *lower sideband* of $U(f)$

It is important to note that either one of the sidebands of $U(f)$ contains all the frequencies that are in $M(f)$

- The frequency content of $U(f)$ for $f > f_c$ corresponds to the frequency content of $M(f)$ for $f > 0$
- The frequency content of $U(f)$ for $f < -f_c$ corresponds to the frequency content of $M(f)$ for $f < 0$
- Hence, the upper sideband of $U(f)$ contains all the frequencies in $M(f)$. A similar statement applies to the lower sideband of $U(f)$

Spectrum of the DSB-SC AM Signal

The other characteristic of the modulated signal $u(t)$ is that it does not contain a carrier component

- As long as $m(t)$ does not have any DC component, there is no impulse in $U(f)$ at $f = f_c$
- That is, all the transmitted power is contained in the modulating (message) signal $m(t)$
- For this reason, $u(t)$ is called a *suppressed-carrier* signal
- Therefore, $u(t)$ is a *DSB-SC* AM signal.

Power Content of DSB-SC Signals

The power content of the DSB-SC signal

$$P_u = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} u^2(t) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} A_c^2 m^2(t) \cos^2(2\pi f_c t) dt$$

$$= \frac{A_c^2}{2} \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} m^2(t) [1 + \cos(4\pi f_c t)] dt \cong \frac{A_c^2}{2} P_m$$

- P_m indicates the power in the message signal $m(t)$
- The last step follows from the fact that $m^2(t)$ is a slowly varying signal and when multiplied by $\cos(4\pi f_c t)$, which is a high frequency sinusoid, the result is a high-frequency sinusoid with a slowly varying envelope, as shown in Figure
- Since the envelope is slowly varying, the positive and the negative halves of each cycle have almost the same amplitude
- Hence, when they are integrated, they cancel each other
- Thus, the overall integral of $m^2(t)\cos(4\pi f_c t)$ is almost zero (Figure 3.6)
- Since the result of the integral is divided by T , and T becomes very large, the second term in Equation (3.2.1) is zero

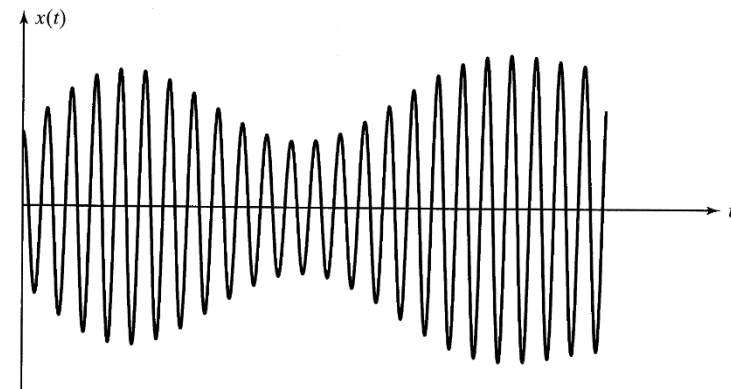


Figure Plot of $m^2(t)\cos(4\pi f_c t)$.

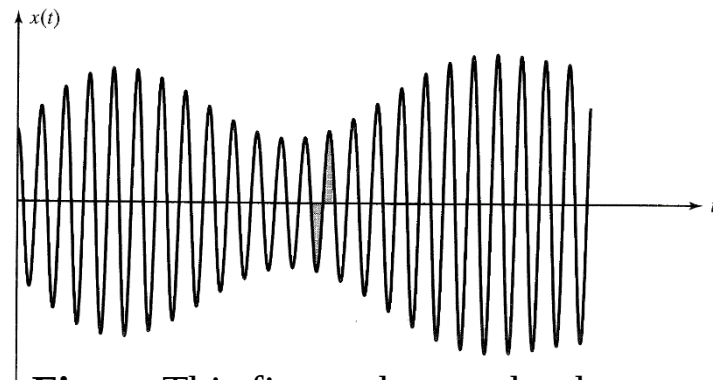


Figure This figure shows why the second term in Equation (3.2.1) is zero.

Demodulation of DSB-SC AM Signals

Suppose that the DSB-SC AM signal $u(t)$ is transmitted through an ideal channel (with no channel distortion and no noise)

$$r(t) = u(t) = m(t)c(t) = A_c m(t) \cos(2\pi f_c t)$$

Then the received signal is equal to the modulated signal,

Suppose we demodulate the received signal by

1. Multiplying $r(t)$ by a locally generated sinusoid $\cos(2\pi f_c t + \phi)$, where ϕ is the phase of the sinusoid
2. We pass the product signal through an ideal lowpass filter with the bandwidth W

3. The multiplication of $r(t)$ with $\cos(2\pi f_c t + \phi)$ yields

$$\begin{aligned} r(t) \cos(2\pi f_c t + \phi) &= A_c m(t) \cos(2\pi f_c t) \cos(2\pi f_c t + \phi) \\ &= \frac{1}{2} A_c m(t) \cos(\phi) + \frac{1}{2} A_c m(t) \cos(4\pi f_c t + \phi) \end{aligned}$$

Demodulation of DSB-SC AM Signals

- The spectrum of the signal is illustrated in Figure 3.7
- Since the frequency content of the message signal $m(t)$ is limited to W Hz, where $W \ll f_c$, the lowpass filter can be designed to eliminate the signal components centered at frequency $\pm 2f_c$ and to pass the signal components centered at frequency $f = 0$ without experiencing distortion
- An ideal lowpass filter that accomplishes this objective is also illustrated in Figure.
- Consequently, the output of the ideal lowpass filter
$$y_l(t) = \frac{1}{2} A_c m(t) \cos(\phi)$$

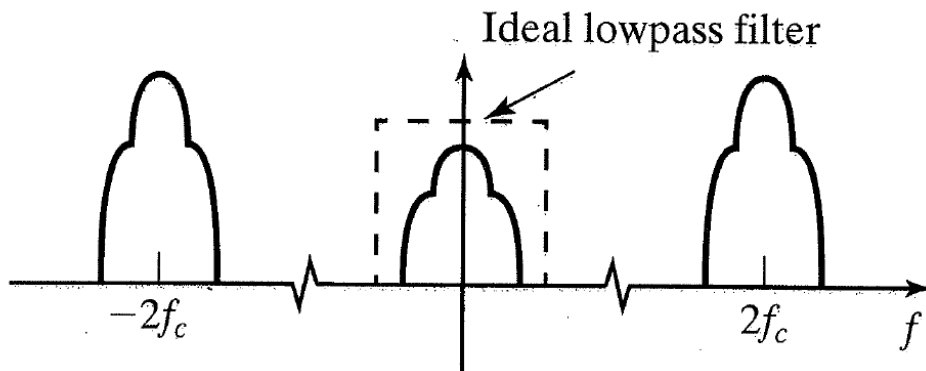


Figure Frequency-domain representation of the DSB-SC AM demodulation.

Demodulation of DSB-SC AM Signals

Note that $m(t)$ is multiplied by $\cos(\phi)$

- Therefore, the power in the demodulated signal is decreased by a factor of $\cos^2 \phi$.
 - Thus, the desired signal is scaled in amplitude by a factor that depends on the phase ϕ of the locally generated sinusoid.
1. When $\phi \neq 0$, the amplitude of the desired signal is reduced by the factor $\cos(\phi)$.
 2. If $\phi = 45^\circ$, the amplitude of the desired signal is reduced by $2^{1/2}$ and the power is reduced by a factor of two.
 3. If $\phi = 90^\circ$, the desired signal component vanishes

The preceding discussion demonstrates the need for a phase-coherent or synchronous demodulator for recovering the message signal $m(t)$ from the received signal

That is, the phase ϕ of the locally generated sinusoid should ideally be equal to 0 (the phase of the received-carrier signal)

Demodulation of DSB-SC AM Signals

A sinusoid that is phase-locked to the phase of the received carrier can be generated at the receiver in one of two ways

- One method is to add a carrier component into the transmitted signal, as illustrated in Figure.
 - We call such a carrier component "a pilot tone."
 - Its amplitude A_p and its power $A_p^2 / 2$ are selected to be significantly smaller than those of the modulated signal $u(t)$.
 - Thus, the transmitted signal is a double-sideband, but it is no longer a suppressed carrier signal

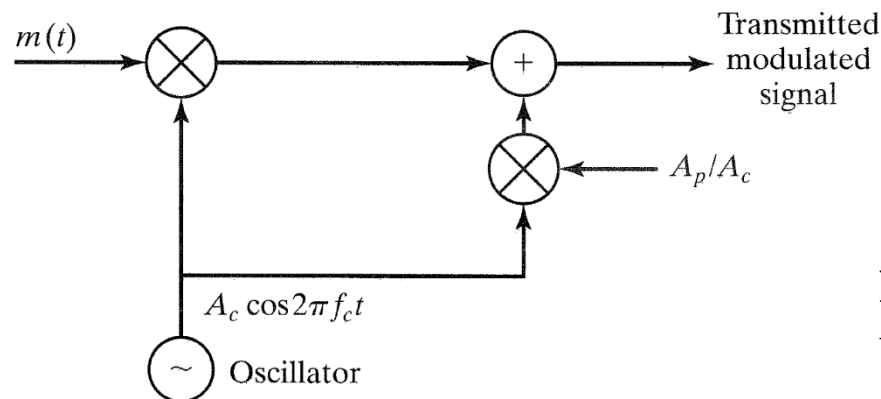


Figure Addition of a pilot tone to a DSB-AM signal.

Demodulation of DSB-SC AM Signals

- At the receiver, a narrowband filter tuned to frequency f_c filters out the pilot signal component
- Its output is used to multiply the received signal, as shown in Figure
- We may show that the presence of the pilot signal results in a DC component in the demodulated signal

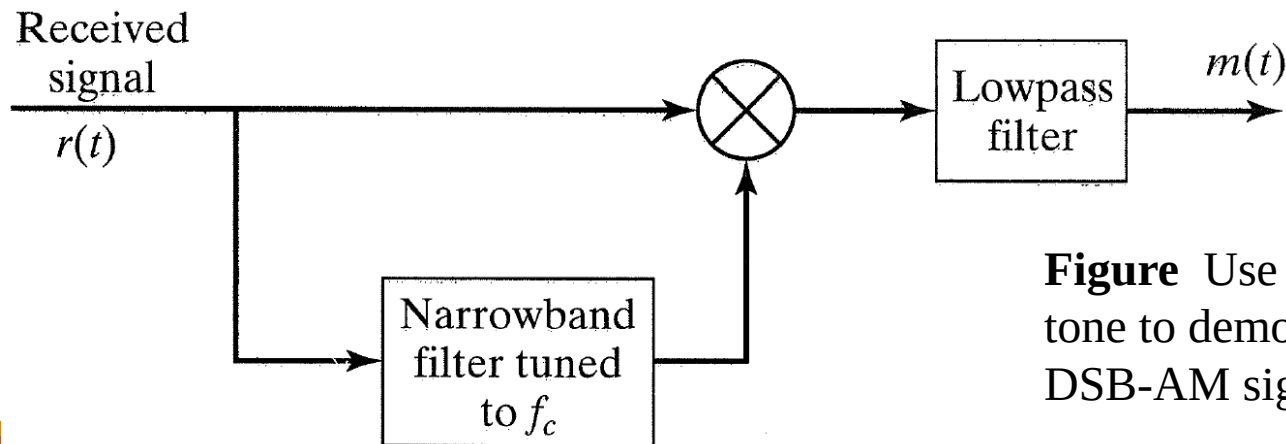


Figure Use of a pilot tone to demodulate a DSB-AM signal.

Demodulation of DSB-SC AM Signals

Adding a pilot tone to the transmitted signal has a disadvantage

- It requires that a certain portion of the transmitted signal power must be allocated to the transmission of the pilot

As an alternative, we may generate a phase-locked sinusoidal carrier from the received signal $r(t)$.

Conventional Amplitude Modulation

A conventional AM signal consists of a large carrier component, in addition to the double-sideband AM modulated signal

- The transmitted signal is expressed mathematically as

$$u(t) = A_c[1 + m(t)]\cos(2\pi f_c t)$$

- The message waveform is constrained to satisfy the condition that $|m(t)| \leq 1$
- We observe that $A_c m(t) \cos(2\pi f_c t)$ is a double-sideband AM signal and $A_c \cos(2\pi f_c t)$ is the carrier component
- Figure illustrates an AM signal in the time domain
- As we will see later in this chapter, the existence of this extra carrier results in a very simple structure for the demodulator
- That is why commercial AM broadcasting generally employs this type of modulation

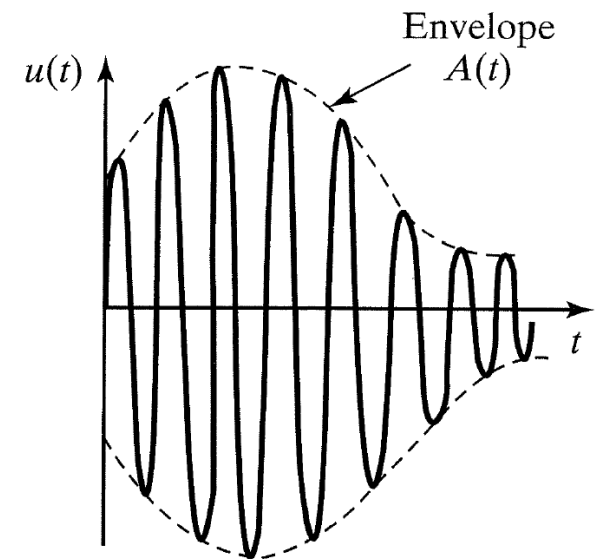


Figure A conventional AM signal in the time domain

Conventional Amplitude Modulation

As long as $|m(t)| \leq 1$, the amplitude $A_c[1 + m(t)]$ is always positive

- This is the desired condition for conventional DSB AM that makes it easy to demodulate, as we will describe

On the other hand, if $m(t) < -1$ for some t , the AM signal is *overmodulated* and its demodulation is rendered more complex

In practice, $m(t)$ is scaled so that its magnitude is always less than unity

- It is sometimes convenient to express $m(t)$ as

$$m(t) = am_n(t)$$

- where $m_n(t)$ is normalized such that its minimum value is -1 and

$$m_n(t) = \frac{m(t)}{\max|m(t)|}$$

- The scale factor a is called the *modulation index*, which is generally a constant less than 1
- Since $|m_n(t)| \leq 1$ and $0 < a < 1$, we have $1 + am_n(t) > 0$ and the modulated signal can be expressed as
- which will never be overmodulated

$$u(t) = A_c[1 + am_n(t)]\cos(2\pi f_c t)$$

Spectrum of the Conventional AM Signal

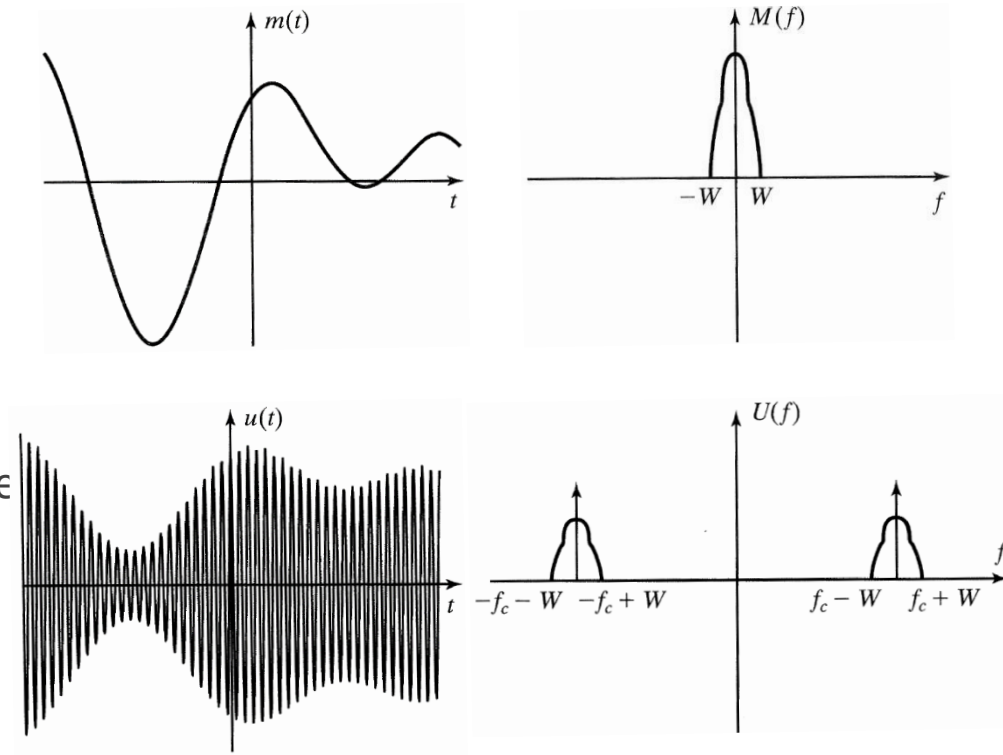
If $m(t)$ is a message signal with Fourier transform (spectrum) $M(f)$, the spectrum of the amplitude-modulated signal $u(t)$ is

$$\begin{aligned} U(f) &= \mathfrak{T}[A_c a_m(t) \cos(2\pi f_c t)] \\ &\quad + \mathfrak{T}[A_c \cos(2\pi f_c t)] \\ &= \frac{A_c a}{2} [M_n(f - f_c) + M_n(f + f_c)] \\ &\quad + \frac{A_c}{2} [\delta(f - f_c) + \delta(f + f_c)] \end{aligned}$$

A message signal $m(t)$, its spectrum $M(f)$, the corresponding modulated signal $u(t)$, and its spectrum $U(f)$ are shown in Figure 3.11

Obviously, the spectrum of a conventional AM signal occupies a bandwidth twice the bandwidth of the message signal

Figure 3.11 Conventional AM in both the time and frequency domain.



Power for the Conventional AM Signal

A conventional AM signal is similar to a DSB when $m(t)$ is substituted with $1 + m_n(t)$

DSB-SC : The power in the modulated signal

$$P_u = \frac{A_c^2}{2} P_m$$

- where P_m denotes the power in the message signal

Conventional AM :

$$P_m = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [1 + am_n(t)]^2 dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [1 + a^2 m_n^2(t)] dt$$

- where we have assumed that the average of $m_n(t)$ is zero
- This is a valid assumption for many signals, including audio signals.

Power for the Conventional AM Signal

Conventional AM,

$$P_m = 1 + a^2 P_{m_n} \longrightarrow P_u = \frac{A_c^2}{2} + \frac{A_c^2}{2} a^2 P_{m_n}$$

- The first component in the preceding relation applies to the existence of the carrier, and this component does not carry any information
- The second component is the information-carrying component
- Note that the second component is usually much smaller than the first component ($a < 1$, $|m_n(t)| < 1$, and for signals with a large dynamic range, $P_{m_n} \ll 1$)
- This shows that the conventional AM systems are far less power efficient than the DSB-SC systems
- The advantage of conventional AM is that it is easily demodulated

Power for the Conventional AM Signal

Efficiency of Conventional AM,

$$E = \frac{\frac{A_c^2}{2} a^2 P_{m_n}}{\frac{A_c^2}{2} + \frac{A_c^2}{2} a^2 P_{m_n}} \times 100 \%$$
$$= \frac{a^2 P_{m_n}}{1 + a^2 P_{m_n}} \times 100 \%$$

Demodulation of Conventional DSB-AM Signals

The major advantage of conventional AM signal transmission is the ease in which the signal can be demodulated

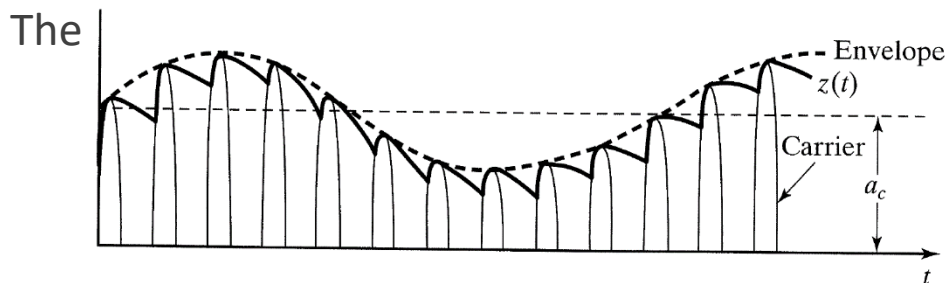
There is no need for a synchronous demodulator

Since the message signal $m(t)$ satisfies the condition $|m(t)| < 1$, the envelope (amplitude) $1+m(t) > 0$

If we rectify the received signal, we eliminate the negative values without affecting the message signal, as shown in Figure.

The rectified signal is equal to $u(t)$ when $u(t) > 0$, and it is equal to zero when $u(t) < 0$

The message signal is recovered by passing the rectified signal through a lowpass filter whose bandwidth matches that of the message signal



is filter is called an *envelope detector*

Figure Envelope detection of a conventional AM signal.

Envelope Detector

As previously indicated, conventional DSB-AM signals are easily demodulated by an envelope detector

A circuit diagram for an envelope detector is shown in Figure 3.27

It consists of a diode and an RC circuit, which is basically a simple lowpass filter

- During the positive half-cycle of the input signal, the diode conducts and the capacitor charges up to the peak value of the input signal
- When the input falls below the voltage on the capacitor, the diode becomes reverse-biased and the input disconnects from the output
- During this period, the capacitor discharges slowly through the load resistor R
- On the next cycle of the carrier, the diode again conducts when the input signal exceeds the voltage across the capacitor
- The capacitor again charges up to the peak value of the input signal and the process is repeated

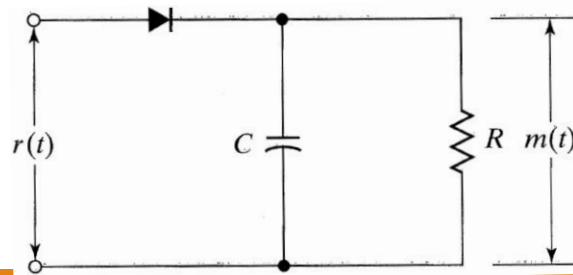


Figure An envelope detector.

Envelope Detector

The time constant RC must be selected to follow the variations in the envelope of the carrier-modulated signal

- If RC is too small, then the output of the filter falls very rapidly after each peak and will not follow the envelope of the modulated signal closely
- This corresponds to the case where the bandwidth of the lowpass filter is too large
- If RC is too large, then the discharge of the capacitor is too slow and again the output will not follow the envelope of the modulated signal
- This corresponds to the case where the bandwidth of the lowpass filter is too small
- Effect of large and small RC values Figure
- For good performance of the envelope detector,

$$\frac{1}{f_c} \ll RC \ll \frac{1}{W}$$
- In such a case, the capacitor discharges slowly through the resistor; thus, the output of the envelope detector, which we denote as $\tilde{m}(t)$, closely follows the message signal

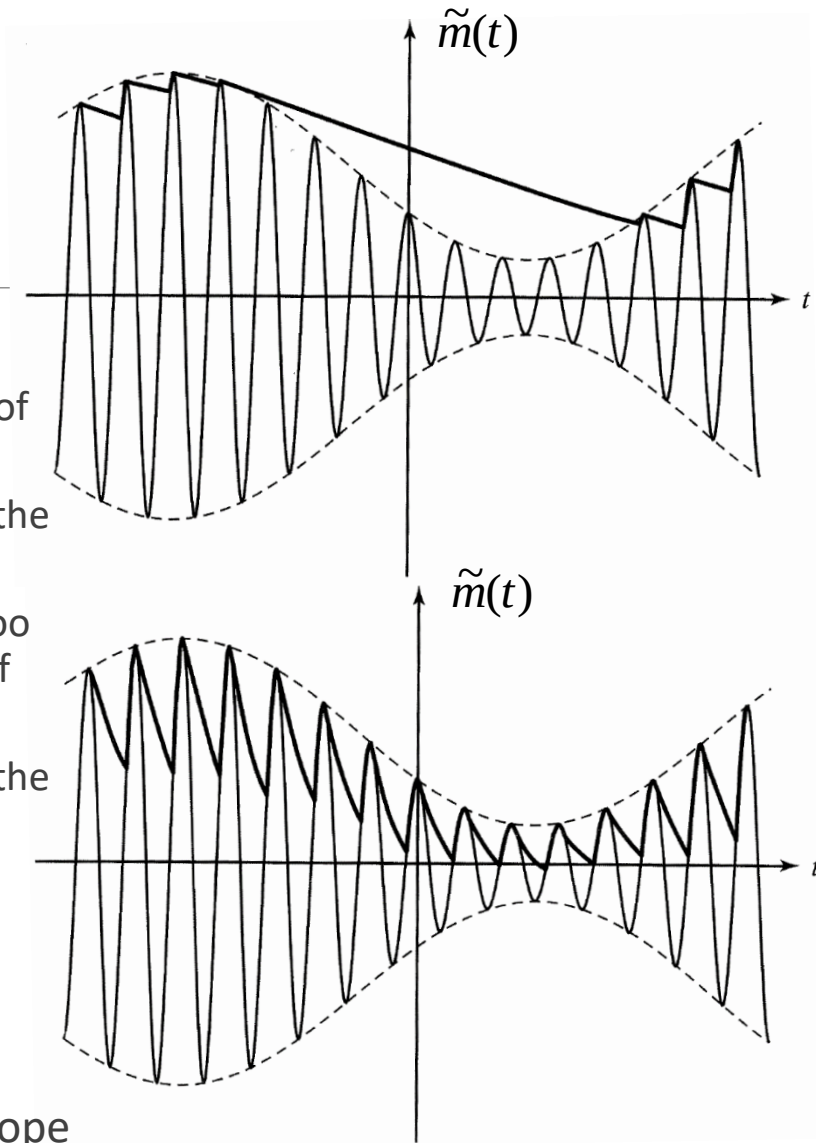


Figure Effect of (a) large and (b) small RC values on the performance of the envelope detector.

Demodulation of Conventional DSB-AM Signals

Ideally, the output of the envelope detector is of the form

$$d(t) = g_1 + g_2 m(t)$$

- where g_1 represents a DC component and g_2 is a gain factor due to the signal demodulator.
- The DC component can be eliminated by passing $d(t)$ through a transformer, whose output is $g_2 m(t)$.

The simplicity of the demodulator has made conventional DSB-AM a practical choice for AM-radio broadcasting

- Since there are literally billions of radio receivers, an inexpensive implementation of the demodulator is extremely important
- The power inefficiency of conventional AM is justified by the fact that there are few broadcast transmitters relative to the number of receivers

Consequently, it is cost-effective to construct powerful transmitters and sacrifice power efficiency in order to simplify the signal demodulation at the receivers

Single-Sideband AM

- A DSB-SC AM signal required a channel bandwidth of $B_c = 2W$ Hz for transmission, where W is the bandwidth of the message signal
- However, the two sidebands are redundant

We will demonstrate that the transmission of either sideband is sufficient to reconstruct the message signal $m(t)$ at the receiver

- Thus, we reduce the bandwidth of the transmitted signal to that of the baseband message signal $m(t)$
- In the appendix at the end of this chapter, we will demonstrate that a single-sideband (SSB) AM signal is represented mathematically as

$$u(t) = A_c m(t) \cos(2\pi f_c t) \mp A_c \hat{m}(t) \sin(2\pi f_c t)$$

- where $\hat{m}(t)$ is the Hilbert transform of $m(t)$ that was introduced in Section 2.6
- The plus or minus sign determines which sideband we obtain
 - The plus sign indicates the lower sideband
 - The minus sign indicates the upper sideband

APPENDIX 3A: DERIVATION OF THE EXPRESSION FOR SSB-AM SIGNALS

Let $m(t)$ be a signal with the Fourier transform (spectrum) $M(f)$

An upper single-sideband amplitude-modulated signal (USSB AM) is obtained by eliminating the lower sideband of a DSB amplitude-modulated signal

Suppose we eliminate the lower sideband of the DSB AM signal, $u_{DSB}(t) = 2A_c m(t) \cos 2\pi f_c t$, by passing it through a highpass filter whose transfer function is given by

$$H(f) = \begin{cases} 1, & |f| > f_c \\ 0, & \text{otherwise} \end{cases}$$

- as shown in Figure 3.16.
- Obviously, $H(f)$ can be written as $u_{-1}(f - f_c) + u_{-1}(-f - f_c)$
- where $u_{-1}(\cdot)$ represents the unit-step function

APPENDIX 3A: DERIVATION OF THE EXPRESSION FOR SSB-AM SIGNALS

Therefore, the spectrum of the USSB-AM signal is given by

$$U_u(f) = A_c M(f - f_c) u_{-1}(f - f_c) + A_c M(f + f_c) u_{-1}(-f - f_c)$$

$$U_u(f) = A_c M(f) u_{-1}(f) \big|_{f=f-f_c} + A_c M(f) u_{-1}(-f) \big|_{f=f+f_c}$$

- Taking the inverse Fourier transform of both sides of Equation (3A.1) and using the modulation and convolution properties of the Fourier transform, as shown in Example 2.3.14 and Equation (2.3.26), we obtain

$$u_u(t) = A_c m(t) * \mathfrak{T}^{-1}[u_{-1}(f)] e^{j2\pi f_c t} + A_c m(t) * \mathfrak{T}^{-1}[u_{-1}(-f)] e^{-j2\pi f_c t}$$

- Next, we note that

$$\mathfrak{T}\left[\frac{1}{2}\delta(t) + \frac{j}{2\pi t}\right] = u_{-1}(f), \quad \mathfrak{T}\left[\frac{1}{2}\delta(t) - \frac{j}{2\pi t}\right] = u_{-1}(-f)$$

- which follows from Equation (2.3.12) and the duality theorem of the Fourier transform

DERIVATION OF THE EXPRESSION FOR SSB-AM SIGNALS

Substituting Equation (3A.3) in Equation (3A.2), we obtain

$$\begin{aligned} u_u(t) &= A_c m(t) * \left[\frac{1}{2} \delta(t) + \frac{j}{2\pi t} \right] e^{j2\pi f_c t} + A_c m(t) * \left[\frac{1}{2} \delta(t) - \frac{j}{2\pi t} \right] e^{-j2\pi f_c t} \\ &= \frac{A_c}{2} [m(t) + j\hat{m}(t)] e^{j2\pi f_c t} + \frac{A_c}{2} [m(t) - j\hat{m}(t)] e^{-j2\pi f_c t} \end{aligned}$$

- where we have used the identities

$$m(t) * \delta(t) = m(t) \qquad m(t) * \frac{1}{\pi t} = \hat{m}(t)$$

- Using Euler's relations in Equation (3A.4), we obtain

$$u_u(t) = A_c m(t) \cos 2\pi f_c t - A_c \hat{m}(t) \sin 2\pi f_c t$$

- which is the time-domain representation of a USSB-AM signal.

DERIVATION OF THE EXPRESSION FOR SSB-AM SIGNALS

The expression for the LSSB-AM signal can be derived by noting that

$$u_u(t) + u_l(t) = u_{DSB}(t)$$

$$A_c m(t) \cos 2\pi f_c t - A_c \hat{m}(t) \sin 2\pi f_c t + u_l(t) = 2A_c m(t) \cos 2\pi f_c t$$

Therefore

$$u_l(t) = A_c m(t) \cos 2\pi f_c t + A_c \hat{m}(t) \sin 2\pi f_c t$$

Thus, the time-domain representation of a SSB-AM signal can generally be expressed as

$$u_{SSB}(t) = A_c m(t) \cos(2\pi f_c t) \mp A_c \hat{m}(t) \sin(2\pi f_c t)$$

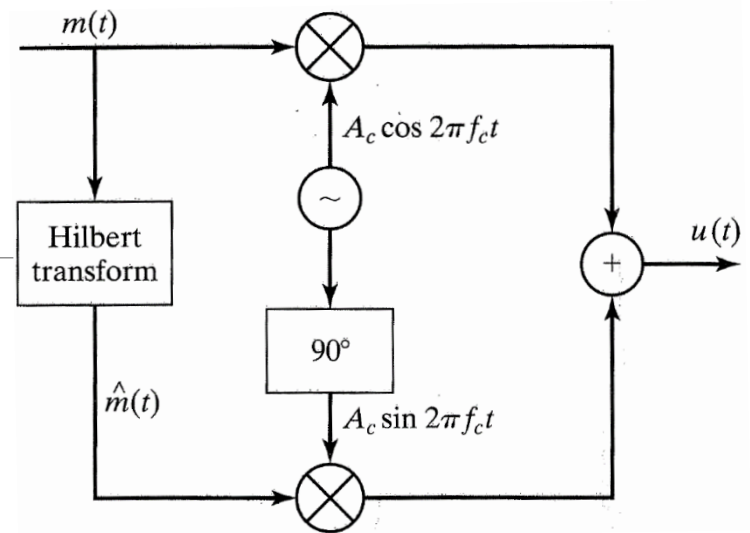
- where the minus sign corresponds to the USSB-AM signal, and the plus sign corresponds to the LSSB-AM signal

Single-Sideband AM

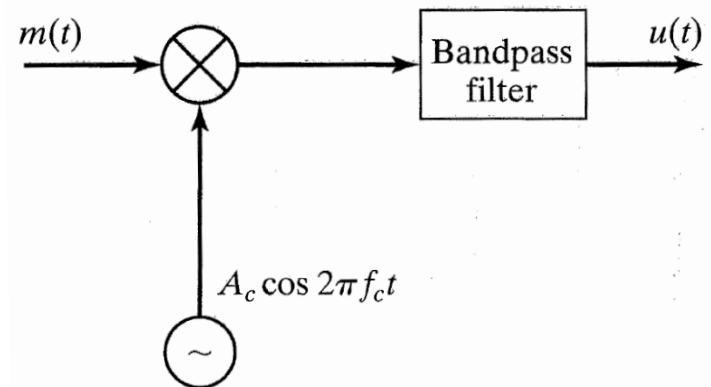
The SSB-AM signal $u(t)$ may be generated by using the system configuration shown in Figure.

- The method shown in Figure. employs a Hilbert-transform filter

Another method, illustrated in Figure, generates a DSB-SC AM signal and then employs a filter that selects either the upper sideband or the lower sideband of the double-sideband AM signal



Generation of a lower single-sideband AM signal.



Generation of a single-sideband AM signal by filtering one of the sidebands of a DSB-SC AM signal.

Demodulation of SSB-AM Signals

To recover the message signal $m(t)$ in the received SSB-AM signal, we require a **phase-coherent** or **synchronous demodulator**, as was the case for DSB-SC AM signals

- For the USSB signal

$$\begin{aligned} r(t) \cos(2\pi f_c t + \phi) &= u(t) \cos(2\pi f_c t + \phi) \\ &= \frac{1}{2} A_c m(t) \cos(\phi) + \frac{1}{2} A_c \hat{m}(t) \sin(\phi) + \text{double frequency terms.} \end{aligned}$$

- By passing the product signal in Equation (3.2.12) through an ideal lowpass filter, the double-frequency components are eliminated, leaving us with

$$y_l(t) = \frac{1}{2} A_c m(t) \cos(\phi) + \frac{1}{2} A_c \hat{m}(t) \sin(\phi)$$

- Note that the phase offset not only reduces the amplitude of the desired signal $m(t)$ by $\cos\phi$, but it also results in an undesirable sideband signal due to the presence of $\hat{m}(t)$ in $y_l(t)$
- The latter component was not present in the demodulation of a DSBSC signal
- However, it is a factor that contributes to the distortion of the demodulated SSB signal

Demodulation of SSB-AM Signals

The transmission of a pilot tone at the carrier frequency is a very effective method for providing a phase-coherent reference signal for performing synchronous demodulation at the receiver

Thus, the undesirable sideband-signal component is eliminated

However, this means that a portion of the transmitted power must be allocated to the transmission of the carrier

The spectral efficiency of SSB AM makes this modulation method very attractive for use in voice communications over telephone channels (wirelines and cables)

In this application, a pilot tone is transmitted for synchronous demodulation and shared among several channels

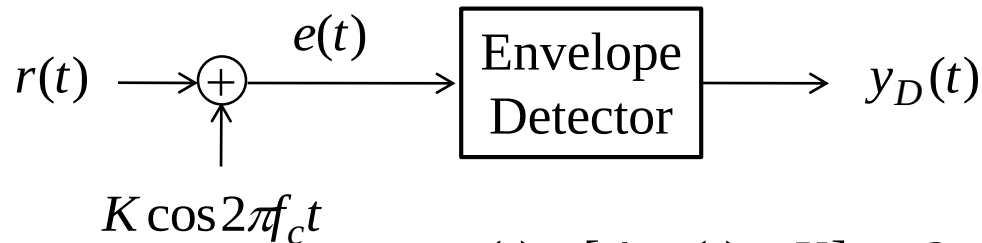
The filter method shown in Figure 3.16, which selects one of the two signal sidebands for transmission, is particularly difficult to implement when the message signal $m(t)$ has a large power concentrated in the vicinity of $f = 0$

In such a case, the sideband filter must have an extremely sharp cutoff in the vicinity of the carrier in order to reject the second sideband

Such filter characteristics are very difficult to implement in practice

Demodulation of SSB-AM Signals

Another method



$$e(t) = [A_c m(t) + K] \cos 2\pi f_c t \mp A_c \hat{m}(t) \sin 2\pi f_c t$$

$$\text{Let } a(t) = A_c m(t) + K \text{ and } b(t) = A_c \hat{m}(t)$$

$$R(t) = \sqrt{a^2(t) + b^2(t)}, \quad \theta(t) = \tan^{-1} \frac{b(t)}{a(t)}$$

$$\begin{aligned} e(t) &= R(t) \cos \theta(t) \cos 2\pi f_c t \mp R(t) \sin \theta(t) \sin 2\pi f_c t \\ &= R(t) \cos(2\pi f_c t \pm \theta(t)) \end{aligned}$$

$$\begin{aligned} \rightarrow y_D(t) &= R(t) = \sqrt{(A_c m(t) + K)^2 + (A_c \hat{m}(t))^2} \\ &\cong A_c m(t) + K, \quad \text{when } K \text{ is large.} \\ &\cong A_c m(t), \quad \text{remove DC component.} \end{aligned}$$

Vestigial-Sideband AM

The stringent-frequency response requirements on the sideband filter in an SSB-AM system can be relaxed by allowing vestige, which is a portion of the unwanted sideband, to appear at the output of the modulator

- Thus, we simplify the design of the sideband filter at the cost of a modest increase in the channel bandwidth required to transmit the signal
- The resulting signal is called ***vestigial-sideband (VSB) AM***
 - This type of modulation is appropriate for signals that have a strong low-frequency component, such as video signals
 - That is why this type of modulation is used in standard TV broadcasting

Vestigial-Sideband AM

To generate a VSB-AM signal, we begin by generating a DSB-SC AM signal and passing it through a sideband filter with the frequency response $H(f)$, as shown in Figure. In the time domain, the VSB signal may be expressed as

$$u(t) = [A_c m(t) \cos 2\pi f_c t] * h(t)$$

- where $h(t)$ is the impulse response of the VSB filter
- In the frequency domain, the corresponding expression is

$$U(f) = \frac{A_c}{2} [M(f - f_c) + M(f + f_c)] H(f)$$

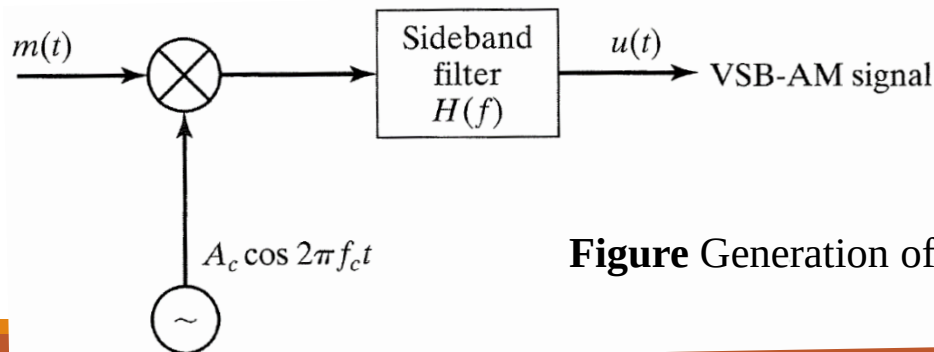


Figure Generation of vestigial-sideband AM signal.

Vestigial-Sideband AM

To determine the frequency-response characteristics of the filter, we will consider the demodulation of the VSB signal $u(t)$

We multiply $u(t)$ by the carrier component $\cos 2\pi f_c t$ and pass the result through an ideal lowpass filter, as shown in Figure

- Thus, the product signal is
$$v(t) = u(t) \cos 2\pi f_c t$$

- or equivalently,
$$V(f) = \frac{1}{2} [U(f - f_c) + U(f + f_c)]$$

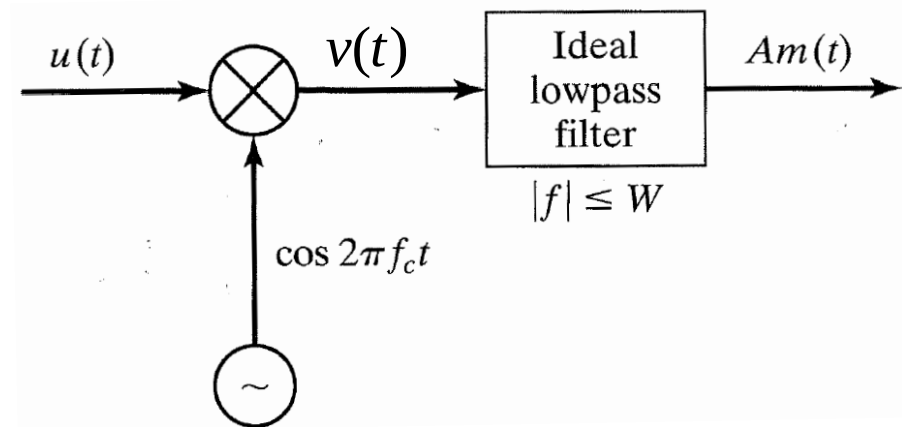


Figure Demodulation of **VSB** signal.

Vestigial-Sideband AM

If we substitute $U(f)$ from Equation (3.2.15) into Equation (3.2.16), we obtain

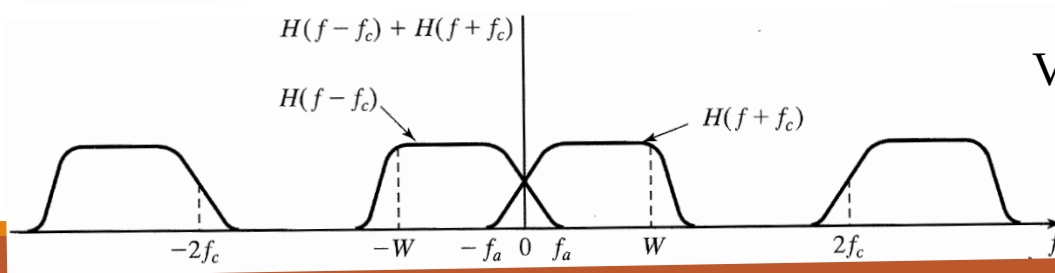
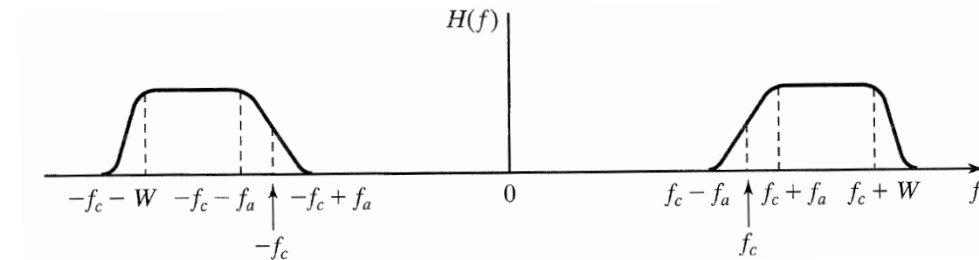
$$V(f) = \frac{A_c}{4} [M(f - 2f_c) + M(f)] H(f - f_c) + \frac{A_c}{4} [M(f) + M(f + 2f_c)] H(f + f_c)$$

- The lowpass filter rejects the double-frequency terms and passes only the components in the frequency range $|f| \leq W$
- Hence, the signal spectrum at the output of the ideal lowpass filter is

$$V_l(f) = \frac{A_c}{4} M(f) [H(f - f_c) + H(f + f_c)]$$

The message signal at the output of the lowpass filter must be undistorted

- Hence, the VSB-filter characteristic must satisfy the condition
- $$H(f - f_c) + H(f + f_c) = \text{constant} \quad |f| \leq W$$



VSB-filter characteristics.

Vestigial-Sideband AM

We note that $H(f)$ selects the upper sideband and a vestige of the lower sideband

It has odd symmetry about the carrier frequency f_c in the frequency range $f_c - f_a < f < f_c + f_a$, where f_a is a conveniently selected frequency that is some small fraction of W , i.e., $f_a \ll W$

Thus, we obtain an undistorted version of the transmitted signal

Figure 3.20 illustrates the frequency response of a VSB filter that selects the lower sideband and a vestige of the upper sideband

In practice, the VSB filter is designed to have some specified phase response

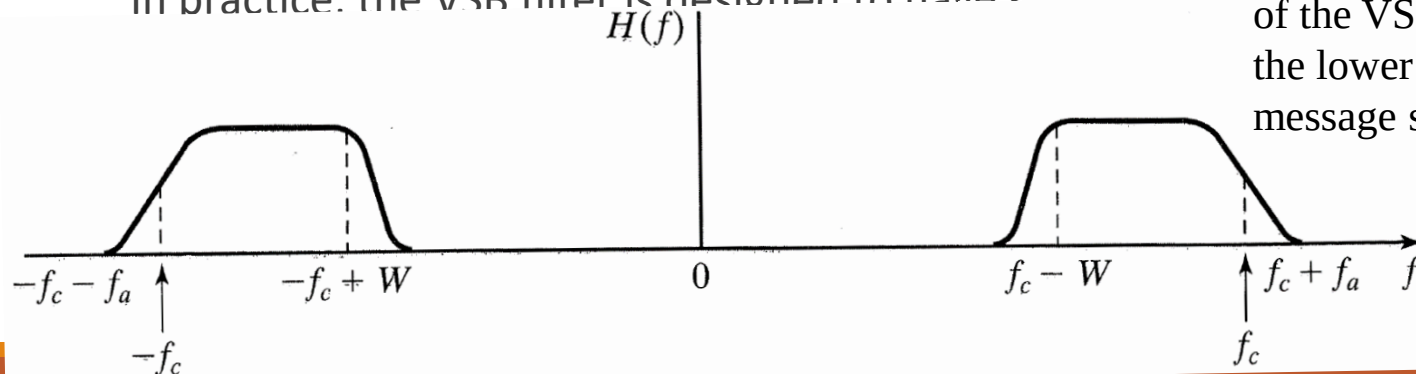


Figure 3.20 Frequency response of the VSB filter for selecting the lower sideband of the message signals.

is a linear

IMPLEMENTATION OF AM MODULATORS AND DEMODULATORS

Power-Law Modulation

$$v_o(t) = a_1 v_i(t) + a_2 v_i^2(t)$$

- where $v_i(t)$ is the input signal, $v_o(t)$ is the output signal, and the parameters (a_1, a_2) are constants

- Then, if the input to the nonlinear device is

$$v_i(t) = m(t) + A_c \cos 2\pi f_c t$$

- Its output

$$v_o(t) = a_1[m(t) + A_c \cos 2\pi f_c t] + a_2[m(t) + A_c \cos 2\pi f_c t]^2$$

$$= a_1 m(t) + a_2 m^2(t) + a_2 A_c^2 \cos^2 2\pi f_c t + A_c a_1 \left[1 + \frac{2a_2}{a_1} m(t) \right] \cos 2\pi f_c t$$

- The output of the bandpass filter with a bandwidth $2W$ centered at $f = f_c$ yields

$$u(t) = A_c a_1 \left[1 + \frac{2a_2}{a_1} m(t) \right] \cos 2\pi f_c t$$

- where $2a_2/m(t)/a_1 < 1$ by design

- Thus, the signal generated by this method is a conventional AM signal

Balanced Modulator

A relatively simple method for generating a DSB-SC AM signal is to use two conventional-AM modulators arranged in the configuration illustrated in Figure.

- For example, we may use two square-law AM modulators as previously described
- Care must be taken to select modulators with approximately identical characteristics so that the carrier component cancels out at the summing junction

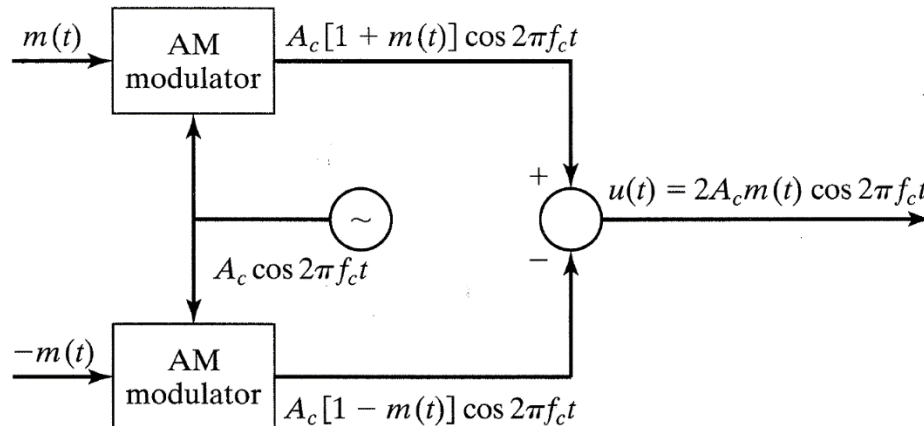


Figure Block diagram of a balanced modulator.

Ring Modulator

Another type of modulator for generating a DSB-SC AM signal is the *ring modulator* illustrated in Figure

- The switching of the diodes is controlled by a square wave of frequency f_c , denoted as $c(t)$, which is applied to the center taps of the two transformers
- When $c(t) > 0$, the top and bottom diodes conduct, while the two diodes in the cross-arms are off
 - In this case, the message signal $m(t)$ is multiplied by $+1$
- When $c(t) < 0$, the diodes in the cross-arms of the ring conduct, while the other two diodes are switched off
 - In this case, the message signal $m(t)$ is multiplied by -1 .
- Consequently, the operation of the ring modulator may be described mathematically as a multiplier of $m(t)$ by the square-wave carrier $c(t)$, i.e.,

$$v_o(t) = m(t)c(t)$$

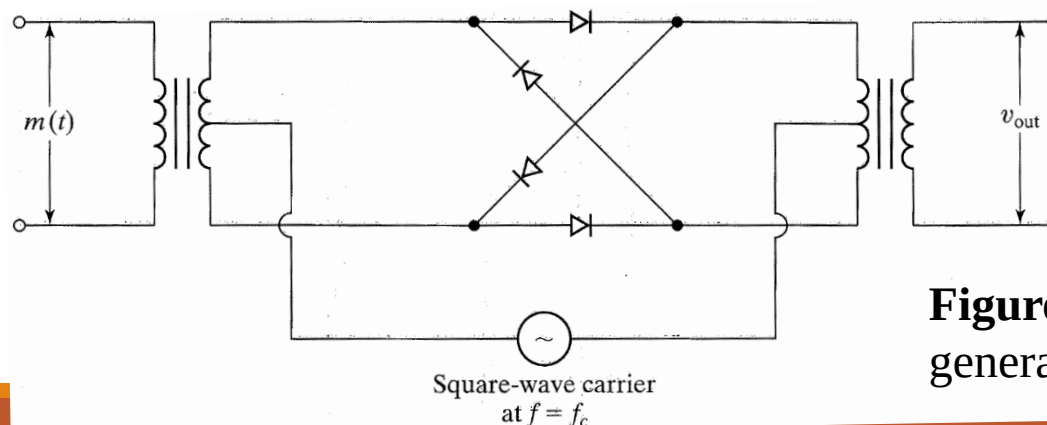


Figure Ring modulator for generating a DSB-SC AM signal.

Ring Modulator

- Since $c(t)$ is a periodic function, it is represented by the Fourier series: p49 ex2.2.2 ㄱ switching modulation

$$c(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cos[2\pi f_c t (2n-1)t]$$

- The desired DSB-SC AM signal $u(t)$ is obtained by passing $v_o(t)$ through a bandpass filter with the center frequency f_c and the bandwidth $2W$

The balanced modulator and the ring modulator systems, in effect, multiply the message signal $m(t)$ with the carrier to produce a DSB-SC AM signal

- The multiplication of $m(t)$ with $A_c \cos(\omega_c t)$ is called a mixing operation
- Hence, a *mixer* is basically a balanced modulator

The method shown in Figure 3.15 for generating an SSB signal requires two mixers

- Two balanced modulators, in addition to the Hilbert transformer
- On the other hand, the filter method illustrated in Figure 3.16 for generating an SSB signal requires a single balanced modulator and a sideband filter

Multiplexing



Frequency Division Multiplexing

FDM

Useful bandwidth of medium exceeds required bandwidth of channel

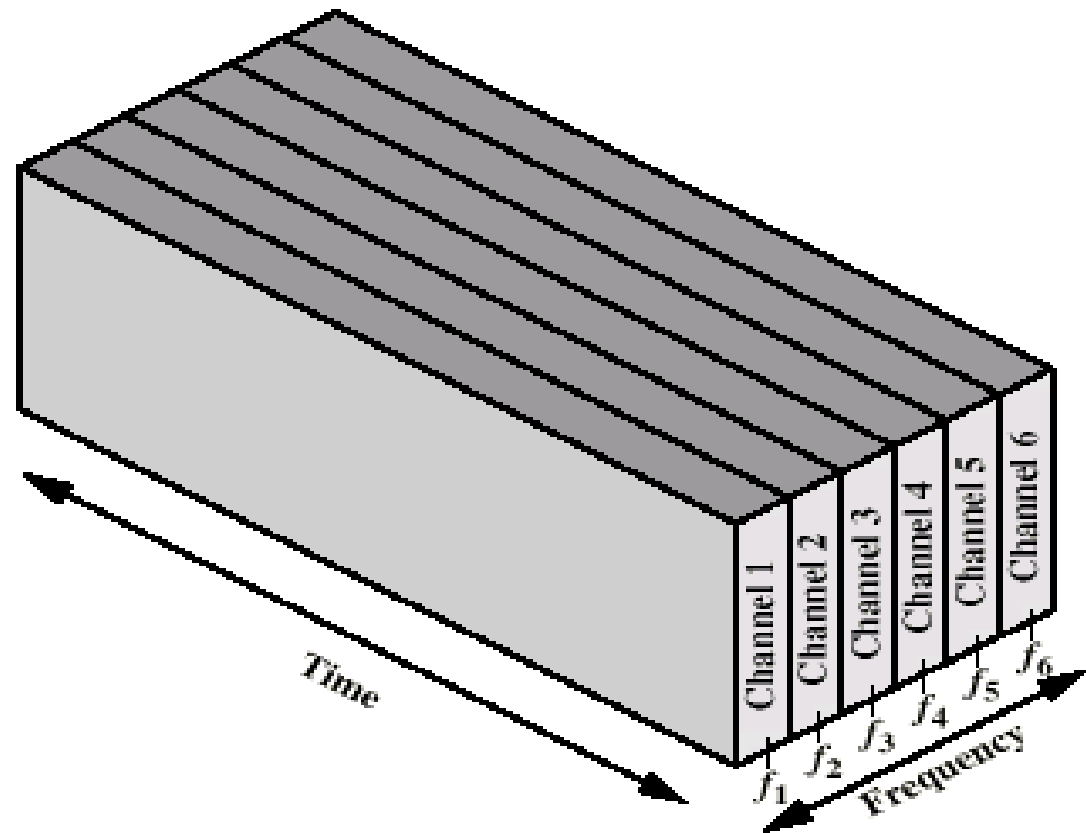
Each signal is modulated to a different carrier frequency

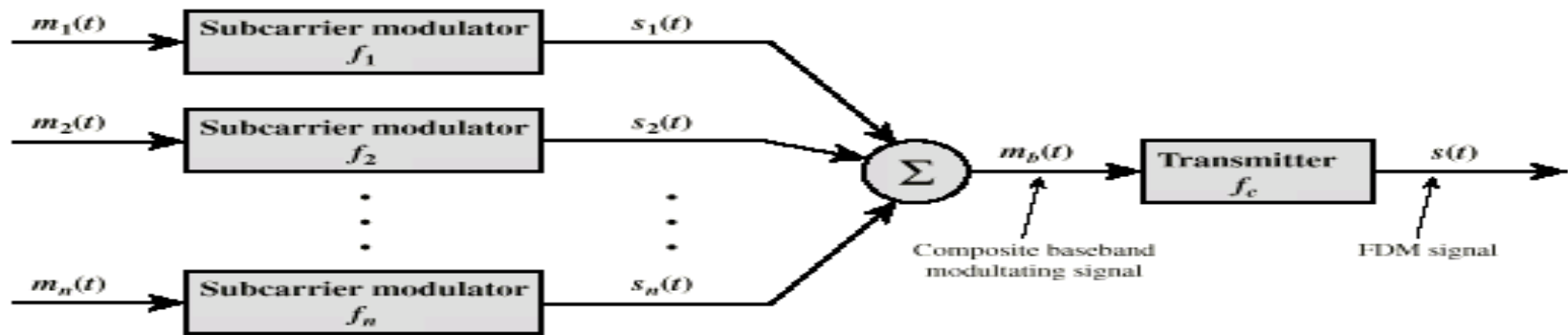
Carrier frequencies separated so signals do not overlap (guard bands)

e.g. broadcast radio

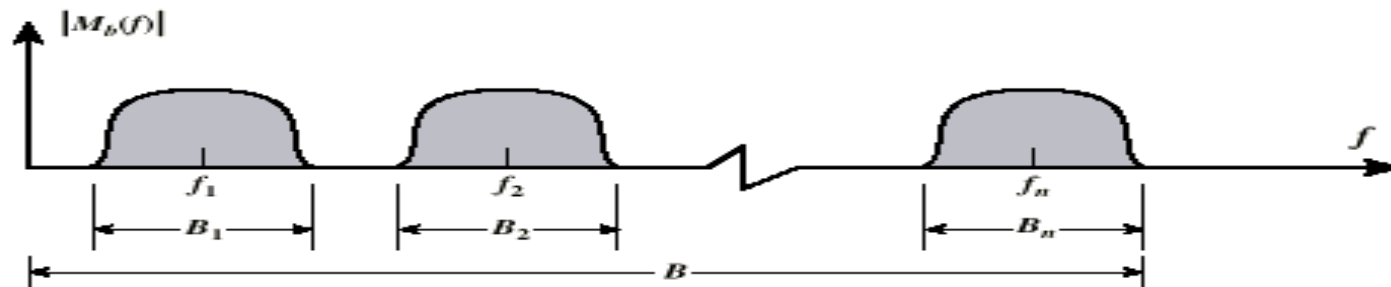
Channel allocated even if no data

Frequency Division Multiplexing Diagram

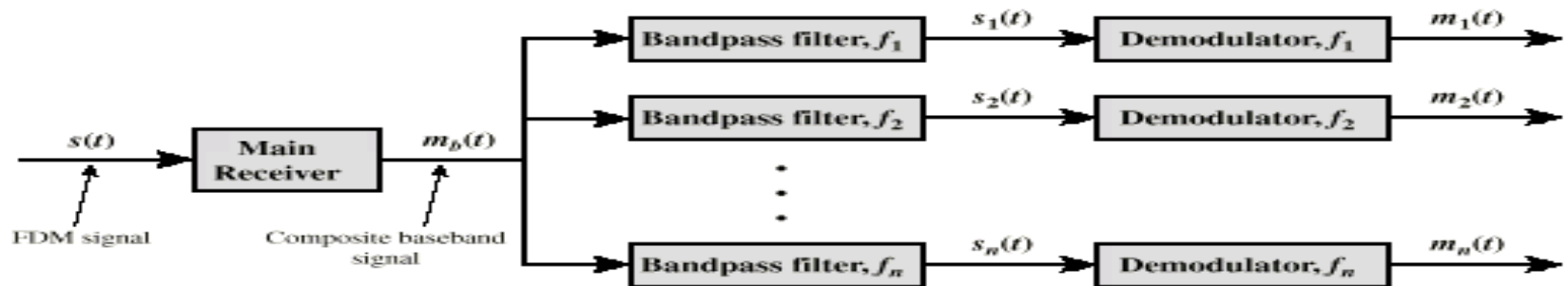




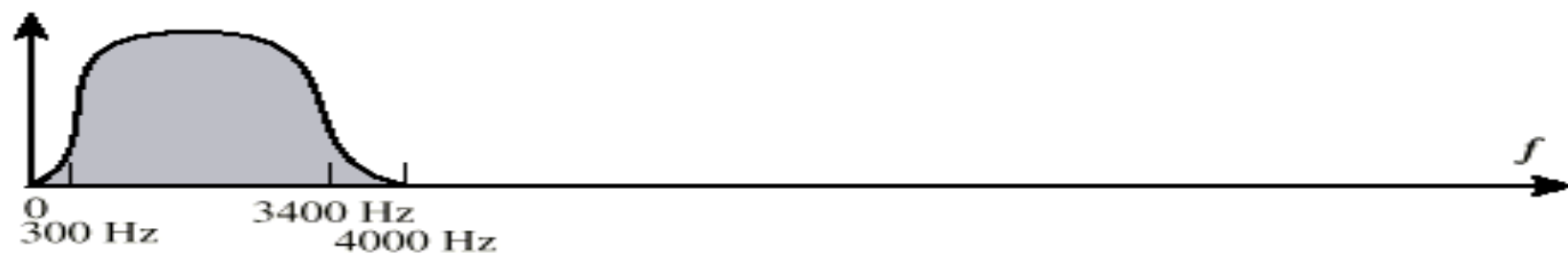
(a) Transmitter



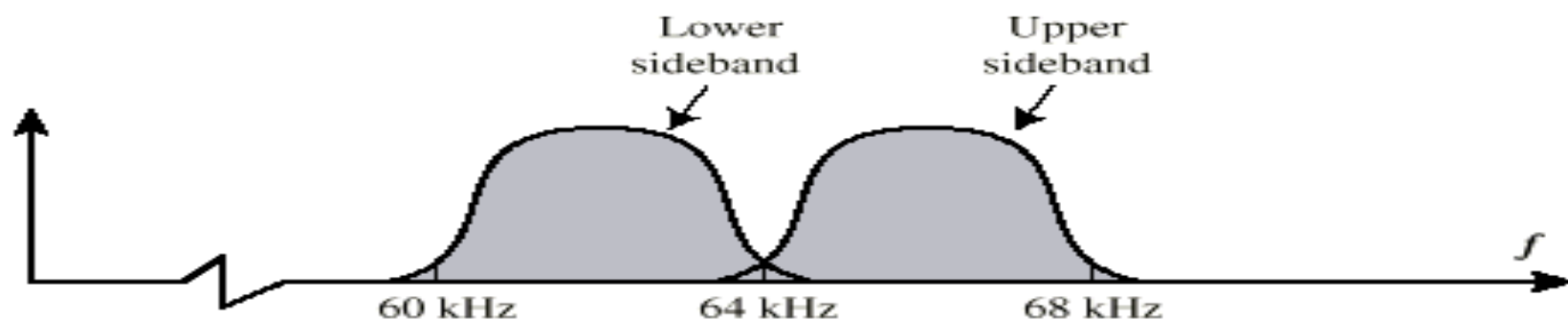
(b) Spectrum of composite baseband modulating signal



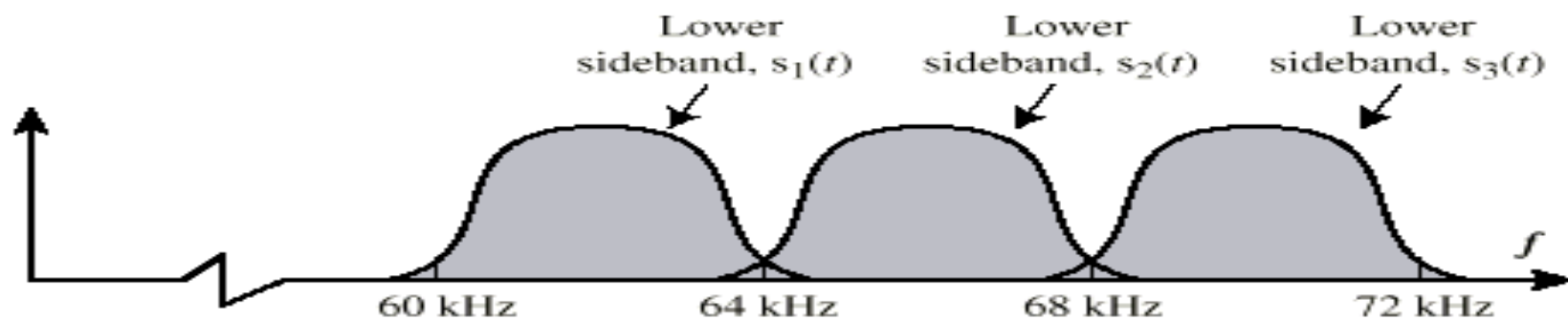
(c) Receiver



(a) Spectrum of $m_1(t)$, positive f



(b) Spectrum of $s_1(t)$ for $f_1 = 64$ kHz



(c) Spectrum of composite signal using subcarriers at 64 kHz, 68 kHz, and 72 kHz